

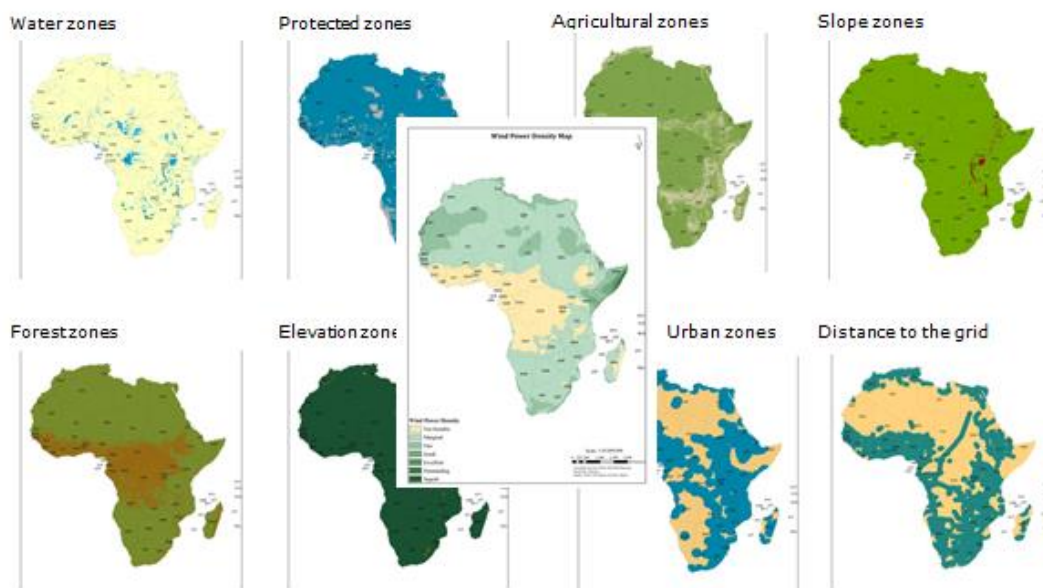


**KTH Industrial Engineering
and Management**

Wind Energy Assessment in Africa

A GIS-based approach

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I. Abstract

This study analyses the potential of onshore wind power on the African continent. Appropriate socio-economic and geographical constraints as well as current technology's efficiencies are applied in order to reach the desired result. The current energy access in Africa is described to illustrate the need of promoting the wind power penetration on the continent. The existing as well as the under construction wind farms are mapped. Thereafter, the methodology of approaching the resource assessment is analyzed. For the energy generation assessment, not only wind speed strength but also its probability of occurrence over a certain period of time is important and thus considered in this study. High resolution wind speed data from Vortex and lower resolution daily wind speed data are combined and processed in order to obtain a fine wind speed distribution and thus wind energy production generated by selected wind turbine models. The different categories of wind power potential are defined and evaluated. Additionally, screening criteria regarding the localization of wind farms are outlined and implemented through GIS analysis. Subsequently interactive maps are prepared. ArcGIS software is used in order to capture, store and manipulate the required data and to obtain a holistic view of the study. The study is conducted at a continental level using a 1km×1km (longitude, latitude) land-use grid as the finest resolution. Ultimately the results of this work are presented and compared with similar approaches and significant conclusions are drawn. Based on the analysis there are some countries that signify high yearly wind energy yield, such as South Africa, Sudan, Algeria, Egypt, Libya, Nigeria, Mauritania, Tunisia and Morocco, whilst Equatorial Guinea, Gabon, Central African Republic, Burundi, Liberia, Benin and Togo indicate the least wind power potential. Also important future work is suggested.

Keywords: *Wind power, Africa, potential, GIS, Renewables mapping*

II. Sammanfattning

Denna studie analyserar potentialen för landbaserad vindkraft på den afrikanska kontinenten. Lämpliga socioekonomiska och geografiska begränsningar samt aktuella vindkraftverkens effektkurvor tillämpas för att nå det önskade resultatet. Den nuvarande tillgången till energi i Afrika beskrivs för att illustrera behovet av att främja vinkraftens penetration på kontinenten. De befintliga vindkraftverken såväl som de under konstruktion kartläggs. Därefter analyseras metoden för att närma sig resurs bedömningen. Bedömningen av energiproduktion och vindhastighet samt dess sannolikhet att inträffa under en viss tid är både viktigt och nödvändigt för denna studie. De olika kategorierna av potential för vindkraftverk definieras och utvärderas. Dessutom beskrivs kriterier av lokalisering för vindkraftverk och genomförs genom GIS-analys. Därefter förbereds interaktiva kartor. ArcGIS software används för att fånga, lagra och manipulera data som krävs samt för att få en helhetssyn av studien. Studien genomförs vid en kontinental nivå genom att använda en 1 km x 1 km (longitud, latitud) rutnät för markanvändning med den finaste upplösningen. Slutligen

presenteras resultaten av detta arbete och jämförs med liknande metoder, viktiga slutsatser dras samt viktigt framtida arbete föreslås.

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The writing of this thesis marks the completion of my Master degree studies and for this reason I would like to express my warm thanks to all members of my family for the unwavering support, understanding and love they gave me during these years. Finally, I thank friends and my fellow students for immediate help and selfless support throughout the period of my studies.

1 Introduction

Driven by concerns over climate change and energy security, the rising cost of fossil fuels, and even economic development interests, wind power is growing worldwide. It is a fact that the cost of wind energy has decreased substantially during the last couple of decades (M. Bolinger et al, 2008). More recently, however, costs have increased due to the rapid and to some extent unexpected growth of the wind industry in the past years (Wiser et al, 2008). Despite the advantages of using wind¹, and the high wind energy potential, it produces just about 1.4% of the world's electric power (IEA, 2009).

The two main barriers to large-scale implementation of wind power are: the apparent intermittency of winds, and the difficulty in identifying decent wind locations, especially in developing countries. The first barrier can be ameliorated by linking multiple wind farms together. Such an approach can virtually eliminate low wind speed events and thus substantially minimize wind power intermittency (C.Archer et al, 2005). The benefits are greater for larger wind power installations areas, as the spatial and temporal correlation of wind speeds is substantially reduced. This study focuses on the second issue, which is the optimal siting of wind farms on the African continent, where it is an emerging need for higher energy production since power is inaccessible, unaffordable and unreliable for most people living there. There are 25 African countries facing an energy crisis and this could be dealt with since Africa is endowed with energy resources, which are yet untapped (The World Bank, 2013).

It is thus very essential to identify and define the amount of wind energy that could be technically exploited in Africa. Maps of wind power potential at 80 m, which is the hub height of a modern wind turbine, will be derived via bilinear interpolation, statistical distribution of wind speed data and implementation of screening criteria related to socio-economic and geographic constraints. The barrier of dealing with regional differences can be overcome by utilizing properly continental Geo-informatic data sets and maps, which are useful tools for resource potential assessment and distribution analysis through GIS.

The analysis of wind speed distribution at a different height than the wind speed is available, is another important factor that should be considered when assessing the wind potentials at a proposed site. The power law extrapolation technique as well as interpolation methods are utilized to obtain estimates of wind speeds at 80 m at each grid cell (grid cell size 1km×1km) given observed wind speeds at 10 m widely available. Results will be used to obtain an estimate of the African theoretical, geographical and technical wind power potential based on state of the art technology.

¹ Wind energy provides many benefits. It is a renewable energy source. Wind power development addresses climate change since it emits very low levels of greenhouse gases. Also it contributes to the reduction of water consumption. Wind power technology experiences high energy intensity compared to other renewables and it is cost competitive to other fuel sources. Furthermore, wind power development leads to decentralized job creation (E.Kondili et al, 2012).

1.1 Wind energy basics

The wind energy is one of the oldest natural resources that were exploited with mechanical systems. The extraction of wind power is an ancient endeavour, beginning with wind-powered ships and wind mills. The last century the wind power technology has been developed and wind turbines are being constructed in order to generate electrical power. The main driver for utilizing wind turbines to generate electricity is the very low CO₂ emissions over the entire life cycle of manufacture, installation, operation and decommissioning and the potential of wind energy to help mitigate the climate change. The stimulus for the enlargement of this field was due to the oil crisis in 70's and the concern over the fossil fuels scarcity (P.Harborne et al, 2009).

The idea is simple. The kinetic energy of the moving air particles can be converted to electricity or energy for pumping water using wind turbines. To understand how wind turbines function, it is useful to briefly consider some of the fundamental facts underlying their operation. The actual conversion process uses the basic aerodynamic force of lift to produce a net positive torque on a rotating shaft, resulting first in the production of mechanical power and then in its transformation to electricity in a generator. Wind turbines, unlike most other generators, can produce energy only in response to the resource (wind) that is directly available. It is not possible to store the wind and use it at a later time. The output of a wind turbine is thus inherently fluctuating and non-dispatchable (Bergeles, 2005). Any system to which a wind turbine is connected must take this variability into account. Another fact is that the wind cannot be transported: it can only be converted where it is blowing. Today, the possibility of conveying electrical energy via power lines compensates to some extent for wind's inability to be transported.

Wind energy derives from the moving air particles across the Earth's surface. All winds are produced by differences in air pressure between two regions. These pressure differences are a result of the uneven heating of the Earth's surface from the sun (Chiras, 2010). It is rather important to understand how much power is available in the wind in order to realize how much energy can be delivered from wind to energy systems. To examine this it is essential to briefly mention how the wind is created. There are several atmospheric forces applied on an air particle to set it into motion and create wind. Such forces are the gravitational force, the pressure gradient force, the Coriolis force, friction, and centrifugal force (Robet Fleagle et al, 1980). This study is not aiming to analyse these forces, but they are described for completeness matters and for understanding the background and the basis of a wind resource assessment, which is no other than the wind itself.

The gravitational force is directed downward perpendicular to the ground and is approximately equal to the mass times the gravitational acceleration ($g \approx 9.8 \text{ m/s}^2$). The pressure gradient force always pushes from higher pressure towards lower pressure. Mathematically, it is written per unit mass as:

$$PGF = -\frac{1}{\text{Air density}} \cdot \frac{\text{Change in pressure}}{\text{Distance}}$$

The Coriolis force (CF) is produced by the rotation of the Earth and it is felt in the frame of reference of the spinning planet. It is expressed as:

$$CF = \pm f \cdot V$$

where $f = 2 \cdot \text{Earth's rotation rate} \cdot \text{the sine of latitude}$

The centrifugal force can be experienced when an object changes its direction of motion even if its speed does not change.

The centripetal acceleration per unit mass can be written mathematically as:

$$CENTF = \frac{V^2}{R}$$

where V is the wind speed and R is the radius of curvature of the curved path that the wind is travelling.

Regarding the frictional force in the atmosphere, it is caused by the flow of wind over the roughness of the Earth's surface. Friction opposes or decelerates the wind in the same way that a rough road surface decelerates a car. Mathematically the frictional force can be written as:

$$FF = -k \cdot V$$

where k is a parameter describing the Earth's surface roughness.

It is accurate to state that the pressure gradient force acts to cause wind, whereas the Coriolis, centrifugal and frictional forces react to the wind to change its speed and/or direction. It is the balance of the above mentioned forces that determine the strength and direction of the wind.

After understanding the basics of how a modern wind turbine works, it is vital to bring up an important question; how much energy is available in the wind and how much can be converted to mechanical energy? The energy available in a wind tube is kinetic energy, which is mathematically expressed as:

$$E = \frac{1}{2} m V^2$$

where m is the mass of the air particles passing through the wind tube and V the wind speed. The mass of the air particles can be obtained from the density definition as:

$$m = \rho \cdot A \cdot l$$

Where ρ is the air density, A is the area of the wind tube; l is the length of the wind tube. The mass flow can be written as:

$$\dot{m} = \frac{dm}{dt} = \rho \cdot A \cdot \frac{dl}{dt} = \rho \cdot A \cdot V$$

The power available in the wind is:

$$P = \frac{dE}{dt} = \frac{1}{2} \cdot \dot{m} \cdot V^2$$

or by inserting the mass flow to the above equation the following formula for the power available in the wind is obtained.

$$P = \frac{1}{2} \cdot \rho \cdot A \cdot V^3$$

This formula is the basis of this analysis in order to obtain the theoretical wind power potential on the African continent.

Can all this kinetic power be converted to mechanical power? Albert Betz, a German physicist showed that the fundamental laws of mass and energy conservation allow no more than 16/27 (59.3%) of the kinetic energy of the wind to be extracted and converted to mechanical energy. This is widely known as the Betz' limit. No wind turbine can produce more than 59.3% of the power available in the wind (Ahmed, 2010). This is due to the fact that air remains in motion after passing through the wind turbine. Betz' limit is the maximum power coefficient that a theoretical wind turbine could reach. The power coefficient describes that fraction of the power in the wind that may be converted to mechanical work. In fact, modern wind turbines reach a power coefficient of about 50% (T.Burton et al, 2011).

1.2 Current energy access in Africa

In comparison with other continents, in Africa nowadays there is a broad shortage of information related to energy matters, such as potential of energy sources, actual installed systems and current energy utilization. This lack of knowledge is even more apparent for renewable energy sources; hence the comparison of the different energy options becomes more complex. However, it is a fact that there is a difficult situation regarding the energy and electricity access on the continent. More than half billion people on the continent lack access to electricity according to African Development Bank (African Development Bank, 2012). The high share of rural population, coupled with the low ability and willingness to pay (affordability), the low energy consumption per capita, as well as the high rate of non-electrified rural areas, have traditionally pushed rural communities to make use of locally available energy sources, mostly biomass from agriculture residues and forest wood for their daily cooking and heating needs (Joint Research Center, 2011).

There is a significant opportunity for the African countries to face the challenge of generating more power to meet the existing and future demand in a clean and sustainable way, since the continent is the most endowed one in renewable energy potential according to the World Energy Council and additionally to that there is no such development and dependence on fossil fuel infrastructure as on continents as Europe and America (World Energy Council, 2010). Based on the above, it can be comprehended that sustainable development could be easier realized in Africa if the renewable potential is to be exploited properly; so as the transition from fossil fuel era to green energy expansion would be based on the sustainability pillars, which are no others but economic, social and environmental sustainability.

1.3 Wind Energy in Africa

While in the past wind energy was considered to be a renewable energy source primarily for developed countries, this is slowly changing. Since 2001, wind energy production has risen significantly worldwide and the cumulative installed global capacity increased from 24 GW at the end of 2001 to 237 GW at the end of 2011 (The Wind Power, 2013). According to the Global Wind Energy Council developing countries and emerging economies - some of whom were barely on the global wind map only a few years ago - have now surged to the forefront in terms of new installed wind capacity (Global Wind Energy Council, 2011).

Also in Africa, the potential of wind energy has started to be recognized and in Egypt, Morocco, Tunisia and South Africa several wind farms have already been installed. Currently, by far the largest share in wind energy production in Africa is held by Egypt and Morocco where most wind power installations are located with total capacities of 550 MW and 1300 MW respectively by the end of 2012 (Global Wind Energy Council, 2011). In Tunisia the annual wind production amounts to about 240 MW² (The Wind Power, 2013). In the graph below it is illustrated how the wind power capacity is expanded in Africa since 1997. A noteworthy growth is experienced from 2006 and on and especially during last year with the total capacity at the end of 2012 exceeding 2 GW. This momentous raise in 2012 is due to the remarkable wind power growth that was experienced in Morocco.

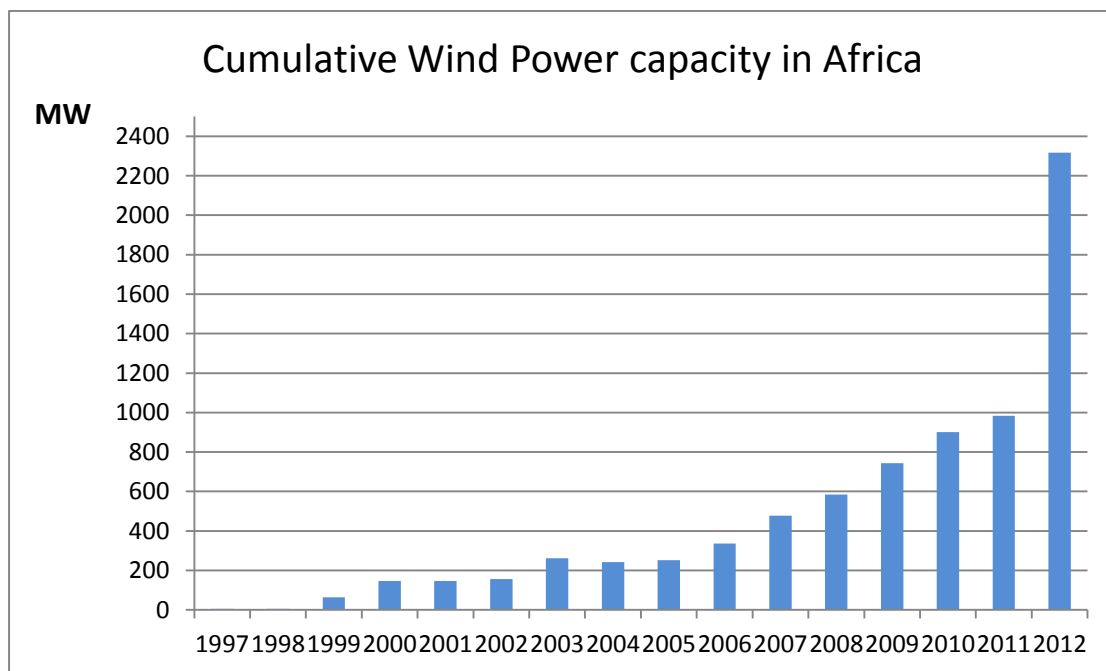


Figure 1.1: Cumulative Wind Power capacity in Africa (The Wind Power, 2013)

However, despite these positive trends and Africa's potential supply of wind energy, installed generation capacity of wind-based electricity located in Africa does not exceed 0,4 percent of global capacity, a fact that makes Africa the laggard continent

² Wind power capacity of 128 MW are under construction in Tunisia

in terms of wind energy development, as it is illustrated in the following pie chart (The Wind Power, 2013). The disparity between potential and extent of exploitation raises questions about barriers to development of wind energy on the continent. The absence of detailed information at individual project level further restricts developers and policy makers' understanding of the market (African Development Bank, 2012).

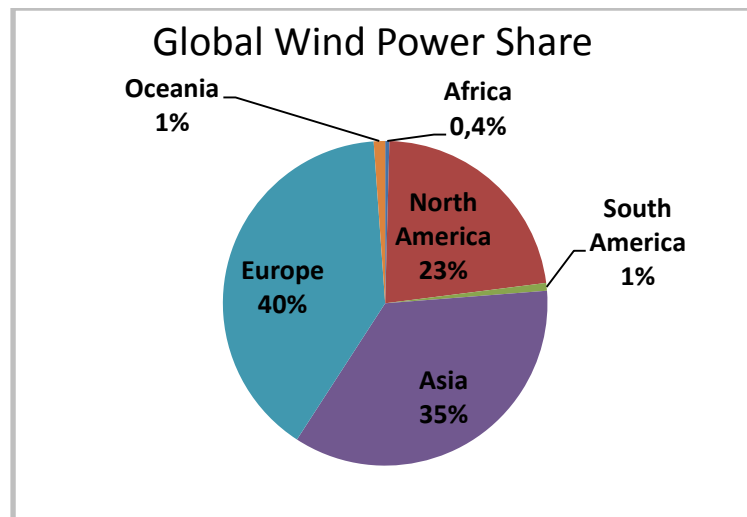


Figure 1.2: Global Wind Power Share (The Wind Power, 2013)

1.4 Literature review

There are several studies performed recently about wind resource estimations using GIS analysis. Archer and Jacobson carry out an evaluation of the global wind power potential and derive wind potential maps at 80m using the Least Square methodology. They quantified theoretical wind power potential maps based on wind speed data (C.Archer et al, 2005). Baban and Parry conducted a study which concerns the siting criteria for wind farms and applied these criteria to an area (40km×40km) in England. Such siting criteria are related to topographical constraints, wind magnitude constraints, population, economic, accessibility, and ecological constraints (Baban Serwan, 2000).

Another substantial study is made in order to distinguish the different categories of wind power potential and how they can be assessed on a global onshore basis. It is described in detail the procedure that starts from the evaluation of the theoretical wind power potential and results to the estimation of the economic potential providing with cost supply curves (M.Hoogwijk et al, 2004). These results are to be compared with this study and are stated in the conclusions.

Furthermore, a remarkable approach regarding the land suitability of a wind farm is introduced by Al-Yahyai et al at their paper Wind farm land suitability indexing using multi-criteria analysis. Several criteria as well as corresponding weighting factors are introduced in this study and applied in a GIS analysis for a case study in Oman (Al-Yahyai et al, 2012). Wind climate modelling using Weibull distribution was presented in the paper of Sarkar et al. It is brought up the importance of such a wind speed distribution to model the wind climate and to estimate properly the wind energy

potential. A case study for Ahmadabad in India is presented to illustrate the methodology followed in this paper and its results (A. Sarkar et al, 2011).

The National Renewable Energy Laboratory published a technical report regarding a GIS method for developing wind supply curves giving information about all the locations than can deliver energy at or below a certain price (K. David et al, 2008). This study considers the case of Zhangbei in China and points out how the wind power density at each grid cell would influence the electricity production cost and hence the required Feed-in tariff.

The aforementioned sources as well as other useful “state of the art” studies are used to develop a solid methodology on how to fulfil the objectives of this project, which are stated in the next paragraph.

1.5 Statement of the objectives

The main goal of this study is to quantify Africa’s onshore wind power potential based on appropriate siting criteria and wind resources. This paper aims to provide estimates of the theoretical, geographical and technical wind power potential in each African country, to indicate possible and sufficient sites to locate wind farms by demonstrating the results in GIS maps and to promote the increase of wind power penetration into the African energy system.

2 Methodology

Initially, data about existing wind farms on the continent and their current production values are collected in order to illustrate the current wind power penetration level and to compare the production values with the estimates obtained from the GIS analysis. Proper siting criteria regarding the wind farms localization are implemented. The siting criteria imply zones, which need to be excluded from the power estimation. Such criteria and the corresponding eliminated zones are discussed subsequently. Thereafter, it is required to derive the technical potential from the theoretical one. Wind speed data as well as land related data are collected and manipulated appropriately in order to obtain the technical wind potential. The data that are used in this study are stated and described in detail in the following paragraph.

Existing wind data including annual, monthly and daily average figures are amassed and treated in order to come up with a wind speed distribution and thus wind power production, based on the power curve of a given turbine model. The flow chart below illustrates how the wind speed data are manipulated until reaching the desired result; i.e. the wind power production at each grid cell. An analysis of the shown process follows.

Accurate information about wind speed is important in determining best sites for wind turbines. The power in the wind is a cubic function of the wind speed, so changes in speed would affect significantly the power production. However, several other data sources are also required to manage to integrate the wind farms on the African continent in a cost effective and social acceptable way. These necessary data are described and their selection is justified.

2.1 Data approach

As it can be seen in the following flow chart, at first the daily wind speed data at 10m height are amassed and treated properly. NASA supplies daily wind speed data at a coarse resolution (grid size $1^{\circ} \times 1^{\circ}$) (NASA, 2012). At this analysis data for year 2012 are processed. Since the wind speed measurements are taken at 10m and the hub height of the proposed wind turbines is at 80m, a proper extrapolation of the wind speed to the hub height should be made.

The wind speed alone is not enough to describe the potential energy from the site. The other important parameter determining the output of a wind farm is the wind speed distribution. This distribution describes the amount of time on a particular site that the wind speed is between different levels (A. Sarkar et al, 2011). This characteristic can be very essential in wind resource assessments, but is often inadequately treated. This distribution is very critical since it is the combination of the power curve of the proposed turbine and the wind speed distribution curve which together determine the energy production.

The wind speed distribution is obtained at 80m for all the grid points. The distribution is described by the Weibull function, which is analysed in chapter 5. Wind speed data from Vortex are provided at a much higher resolution at 80m. However, these data consist of annual mean wind speeds at each grid point. The idea is to combine the

wind speed distribution obtained from the lower resolution daily data and the annual mean wind speed of the higher resolution.

By doing so, a more appropriate distribution for analysing the technical wind power potential can be obtained. This can be done by applying the method of bilinear interpolation, which is described below.

Wind power density is also calculated at each grid cell and presented in a GIS map. This gives an insight of the theoretical wind power potential, while the geographical potential arises by introducing wind farm related siting criteria. The next step is to combine the wind speed distribution with the power curve of 2 selected wind turbine models and simultaneously apply suitable availability factors and potential operating hours of wind turbine. Spacing factors are applied eventually through GIS in order to conclude to a finer estimation of the wind power potential that could be technically exploited. All the aforementioned steps are described in detail in the following paragraphs.

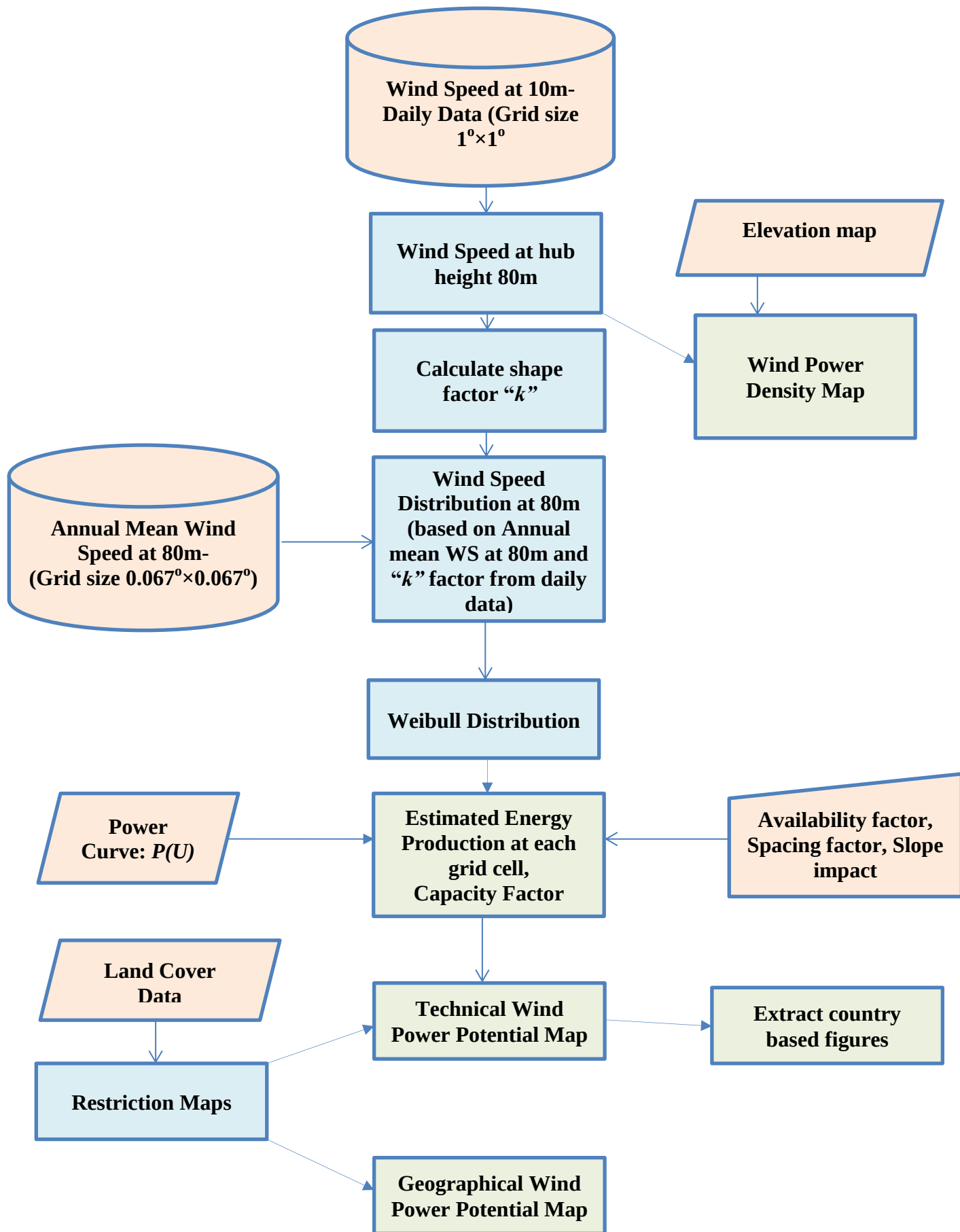


Figure 2.1: Flow chart-methodology

2.2 Data description

In order to reach an accurate and reliable estimation of the wind power potential on the continent of Africa, proper data should be collected, modified and analyzed. Several sources are utilized not only for the wind speed data, but also for administrative country data, elevation and slope figures, population centers, land cover information and protected areas. Wind speed data are used in order to define the wind power potential, while the rest to narrow down from theoretical potential to the technical one, since the latter includes certain limitations and obstacles as discussed in the methodology section. The data sources used for the completion of the project are presented in the following paragraphs.

2.3 Cartography of the existing wind farms

In the next table data regarding existing wind farms in Africa are collected. More specifically, the official name, the installed capacity and the corresponding estimated annual production, as well as the localization³ of each wind farm are presented below (The Wind Power, 2013). The latter will be considered in the analysis that follows in the ArcGIS approach in order to place the current wind power capacity where it is actually installed and to avoid overlapping areas that are already been utilized for wind energy production. Also, the data below will be used to validate the results of the technical wind power potential. A map with the existing and under construction wind farms and the technical wind power potential is presented in the Results section (Figure 5.7).

Table 2.1: Country based data for wind farms in Africa

Countries/ Wind Farms	Total Nominal Power (MW)	Estimated Annual Production (GWh)⁴	Latitude	Longitude
Algeria	24.2			
Kabertene in Adrar	24.2 ⁵	60	28° 26' 59.9"	-0° 4' 0"
Cape Verde	38			
Boa Vista Island	2.5	6	16° 8' 14.1"	-22° 51' 29.5"
Cabeolica	22.5	56	16° 8' 14.1"	-22° 51' 29.5"
Mindelo	0.9	2	16° 52' 12"	-24° 58' 11.9"

³ The geodetic system is WGS84

⁴ The annual production is estimated for an equivalent of 2,500 hours of full load/year

⁵ 10,2 MW under construction

Praia	0.9	2	14° 55' 11.9"	-23° 31' 48"
Sal	0.6	1	16° 43' 11.9"	-22° 55' 12"
Egypt	544.82			
Zarafana 1	30	75	29° 6' 35.9"	32° 39' 35.9"
Zarafana 2	33	82	29° 6' 35.9"	32° 39' 35.9"
Zarafana 3	30.36	75	29° 6' 35.9"	32° 39' 35.9"
Zarafana 4	46.86	117	29° 6' 35.9"	32° 39' 35.9"
Zarafana 5	85	212	29° 6' 35.9"	32° 39' 35.9"
Zarafana 6	79.9	199	29° 6' 35.9"	32° 39' 35.9"
Zarafana 7	119.85	299	29° 6' 35.9"	32° 39' 35.9"
Zarafana 8	119.85	299	29° 6' 35.9"	32° 39' 35.9"
Eritrea	0.825			
Assab	0.825	2	13° 0' 7"	42° 44' 17.4"
Ethiopia	171.18			
Adama	51	127	8° 32' 23.8"	39° 16' 19.1"
Ashegoba at Mekele	120.18	300	N/A ⁶	N/A
Gambia	0.15			
Batakunku	0.15	0.375	13° 19' 26.3"	-16° 48' 0.6"
Kenya	18.7			
Ngong	13.6	34	-1° 21' 23"	36° 39' 28.6"
Ngong Hills	5.1	12	-1° 21' 23"	36° 39' 28.6"
Libya	20			
Libya Wind Farm	20	50	N/A	N/A
Mauritania	35.9			
Nouadhibou	4.4	11	20° 55' 59.9"	-17° 1' 59.9"
Nouakchott	31.5	78	18° 5' 2.6"	-15° 58' 42.3"
Mauritius	1.1			
Rodrigues Island	1.1	2	-19° 42' 19"	63° 27' 10.4"
Morocco	1159.2			
Akhfainir	101.87	254	35° 49' 12"	-5° 27' 0"

⁶ N/A indicates not available data

Al Koudia Al baida	53.9	134	35° 49' 12"	-5° 27' 0"
Ciments du Maroc	5	12	27° 9' 12.9"	-13° 12' 11.9"
Foud El Wad	50.6	126		
Foum El Oued Farm	200	500	28° 25' 59.9"	-11° 5' 59.9"
Haouma	55	137		
Laâyoune Farm	51.1	127	27° 9' 44.4"	-13° 12' 5.8"
Sendouk Farm	65	162	35° 46' 0"	-5° 47' 59.9"
Tan Tan	10	25	28° 25' 22.1"	-11° 5' 21.5"
Tanger	140.25	350	35° 46' 48"	-5° 48' 35.9"
Tarfaya	300	750	27° 56' 8"	-12° 55' 7.3"
Tarfayer	65	162	31° 30' 36"	-9° 44' 24"
Tetouan	32	80	35° 34' 16.9"	-5° 22' 20.3"
Tetouan II	10.2	25	35° 34' 16.9"	-5° 22' 20.3"
YNNA Bio Power	20	50	31° 30' 32.5"	-9° 44' 50.8"
Mozambique	0.3			
Praia da Rocha windfarm	0.3	0.75	-23° 55' 22"	35° 31' 35"
Namibia	0.22			
ErongoRED	0.22	0,55	-22° 57' 0.9"	14° 30' 37"
Nigeria	10.175			
Federal Ministry of Energy wind farm	10.175	25	12° 59' 34.8"	7° 36' 22.1"
South Africa	61.21			
COEGA	43	107	-33° 55' 57.5"	25° 34' 11.8"
Darling	1.25	33	-33° 22' 48"	18° 22' 47.9"
Klipheuwel	3.16	7	-33° 41' 23.9"	18° 43' 48"
Nelson Mandela Bay Stadium	1.8	4	-33° 55' 57.5"	25° 34' 11.8"
Tanzania	50			
Njiapanda	50	125	-8° 59' 3.7"	33° 37' 18.3"

Tunisia	242.36 ⁷			
Centrales éoliennes de Bizerte – 1/Kchabta	59.4	148,5	37° 9' 24.3"	9° 46' 51.3"
Centrales éoliennes de Bizerte – 1/Metline	60.72	151,8	37° 15' 0"	10° 3' 0"
Centrales éoliennes de Bizerte – 2/Kchabta	34.32	85,8	37° 9' 24.3"	9° 46' 51.3"
Centrales éoliennes de Bizerte – 2/Metline	34.32	85,8	37° 15' 0"	10° 3' 0"
Sidi Daoud	53.6	134	37° 3' 53.7"	11° 2' 18.8"

2.4 Wind Speed Data Collection

Wind data are made available for this study by several sources. VORTEX provided the Division of Energy Systems Analysis with a high resolution “annual - wind-speed map” of the African continent with a grid size of approximately 6km×6km to 7km×7km depending on the exact location on the earth, or 0.067°×0.067°. Wind speed was provided for a height of 80m above ground (Vortex, 2012)⁸. Another data source for wind speed distribution is the NASA Climatology Resource for Agroclimatology. Daily wind speed data are being provided at 10m height from January 1983 till present time (NASA, 2013). These high resolution data are used to validate the rest wind speed data used throughout the analysis⁹. Furthermore, the National Aeronautics and Space Administration (NASA) delivers an open source for monthly and annual averaged wind speed values for a 10-year period (July 1983-June 1993) at 50m above the surface of the earth (NASA, 2012). Besides the previously mentioned sources, reliable wind speed data from sounding stations in year 2000, which record twice a day, are provided by Prof. Cristina L. Archer and Prof. Mark Jacobson (C.Archer et al, 2005). The last two data sources are suggested for future use and may be useful for validating the data analysis made in this study.

2.5 Administrative Country data

⁷ The wind farms Centrales éoliennes de Bizerte – 1,2/Kchabta and 2/Metline are under construction.

⁸ Disclaimer: The data from Vortex cannot be further distributed.

⁹ Christina Archer is Associate Professor in the College of Earth, Ocean, and Environment at University of Delaware and Mark Jacobson is Professor in Department of Civil and Environmental Engineering at Stanford University. They published a paper in 2005 which concerns the Evaluation of Global Wind Power.

The boundaries between the African countries should be clearly stated, so as to determine the wind power potential for each one of them. Freely available country boundaries data for Africa are used in order to define country areas. Data include exact boundary locations as well as country sizes and population levels. The data source is United Nation Second Administrative Level Boundaries data set project (UNSalB, 2001) and the Global Administrative database (GADM, 2012).

2.6 Digital Elevation and Slope Data

Digital elevation and slope data are used to define certain land areas that are excluded from the analysis – exclusion parameters include a certain height over sea level but also steep slopes which make the application of large scale installations impossible. Digital elevation data are extracted from so called digital elevation models which offer a 3D-representation of a terrain's surface. Freely available digital elevation data provided by the Consultative Group on International Agricultural Research (CGIAR, 2013) are used in the analysis as well as other freely available sources for elevation and slope data (though at lower resolution levels) in GIS format such as Global Agro Ecological Zoning Model constructed by the International Institute for Applied Systems Analysis (IIASA, 2013).

2.7 Socioeconomic data

Available maps which include all population centres with more than 50,000 inhabitants are also used and analysed. Socioeconomic Data and Applications Centre offer such valuable data (SEDAC, 2012). Such maps are helpful to define areas which are extremely rural and potentially not suitable for large scale electricity production. In the land exclusion analysis, a scenario is created; all areas further away than 200km from any settlement (of more than 50,000 inhabitants) are to be excluded. The European Joint research centre provides map data on population centres as well as travel times to these centres as a measure of remoteness (JRC, 2009).

2.8 Land Cover Data

Land cover is separated into 20 categories including different types of forest, shrub land, herbaceous plants, crop land, paddy fields, mangroves, wetlands, urban areas, snow/ice and water bodies (ISCGM, 2012)¹⁰. Land Cover maps are also available at the Joint Research Centre database.

The European Joint research centre provides data map on African protected areas (JRC, 2013). This data map contains information on 741 protected areas, across 50 countries, and includes information on 280 mammals, 381 bird species and 930 amphibian species, and a wide range of climatic, environmental and socioeconomic information. It is rather important to consider the protected areas in the exclusion

¹⁰ The data are provided in TIFF format with a world file for geo-referencing and with a 30 arc seconds resolution.

criteria in order to shield the natural wealth of the continent and sustain the biodiversity values.

3 Data analysis

In this chapter it is depicted how the collected required data are analysed and processed in a convenient way for the subsequent GIS analysis. Before this comprehensive data analysis, it is essential to define the different wind power potential categories so as to gain an overview of the most significant parameters that play a key role to the wind power utilization. Furthermore, these parameters are used further in this study to reach the objective of estimating the technical wind power potential.

3.1 Wind Power Potential categories

The distinction between the different potential categories was stated by van Wijk and Coelingh in 1993 (W.Coelingh, 1993). As displayed in the following table each category narrows down the previous one, since it includes certain limitations and obstacles.

Table 3.1: Overview of the different potential categories

Potential Category	Description
Theoretical Potential	The total energy content of the wind (kWh/annum)
Geographic Potential	The total amount of land area available for wind turbine installation taking geographical constraints into account (km ²)
Technical Potential	The wind power generated at the geographical potential including energy losses due to the power density of the wind turbines and the process of generating electricity using wind turbines (kWh/annum)
Economic Potential	The technical potential that can be realized economically given the cost of alternative energy sources (kWh/annum)
Implementation Potential	The amount of economic potential that can be implemented within a certain timeframe, taking (institutional) constraints, specific legislation framework and incentives into account (kWh/annum)

It can be clearly noticed that the separate categories are not strictly defined and may be interpreted in different ways. However, the sequence included in the categories facilitates the study of the constraints that reduce the wind power potential and gives a thorough insight of the important factors affecting it.

To analyse the implementation potential, one needs quantification of the electricity system and of important social values and institutional interventions such as subsidies,

tolerated investment risks, local preferences on landscape, etc. These cannot be evaluated unless one defines a specific quantitative scenario based on population and economic dynamics. Possible barriers to implementation are visual and financial constraints or competition with other power generation options. However, it should be noticed that various social factors may already be encountered when estimating the geographical and technical potential. The investigation of the implementation potential is out of the scope of this analysis.

3.2 Theoretical potential

At first the theoretical potential needs to be calculated. At grid cell level, it is conceptually difficult to calculate the power in the wind. Instead, a measure of wind energy intensity is enumerated. The wind contains an amount of kinetic energy that can be expressed in terms of the mass of the air and the speed of this mass. The kinetic potential from wind energy (also known as wind power density) is a useful way to evaluate the wind resource available at a potential site and is expressed as:

$$WPD = \frac{1}{2} \rho U^3$$

where WPD is the power in W per m² swept area ¹¹, ρ is the air density in (kg/m³), and U is the wind speed in (m/s) at each grid point (J. Rogers et al, 2009). WPD stands for wind power density. If however there are time series for the wind speed data, the wind power density is a nonlinear function of the probability density function of wind velocity and air density and is expressed as follows:

$$WPD = \frac{1}{2} \cdot \iint \rho U^3 f(\rho, U_j) d\rho dU$$

which can be simplified to the following formula since the number of known wind speed data are limited to 366 for each grid point

$$WPD = \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \rho_j \cdot U_j^3$$

where n is the number of wind speed readings and ρ_j and U_j are the j^{th} (1st, 2nd, 3rd, etc.) readings of the air density and wind speed respectively at a particular site (Jie Zhang et al, 2012). This entails a calculation for every data interval since the air density and wind speed change with every data reading. This formula can be generalized for all the grid points. The air density is assumed to be the same at a certain location. Thus the above expression can be written as follows for the WPD at k -th grid point.

$$WPD_k = \frac{1}{2} \cdot \frac{1}{n} \cdot \rho_k \sum_{j=1}^n U_j^3$$

¹¹ Swept area is the area covered by the wind turbines blades and given by the formula $\frac{\pi d^2}{4}$, where d is the rotor diameter.

The air density¹² can be determined with the following formula which takes into account the location's elevation above sea level $-z$ -in (m) (Hughes, 2000). This formula approximates more precisely the air density and hence the wind power density. It gives a good long-term average value of air density.

$$\rho = 1.225 - (1.194 \cdot 10^{-4}) \cdot z$$

Air density is inversely related to elevation and temperature. It decreases with increasing elevation or increasing temperature. The above formula neglects temperature variation. It would have been even more precise and logical to consider the temperature variation as well. However, it can be easily shown that even large temperature differences would result in small WPD differences. Hence, the above simplification is fair enough for an early stage approach of the wind resource.¹³

The ultimate expression to estimate the WPD at each grid cell is the following:

$$WPD_k = \frac{1}{2n} \cdot [1.225 - (1.194 \cdot 10^{-4}) \cdot z_k] \sum_{j=1}^n U_j^3$$

No siting criteria, power coefficients and spacing factors are considered at this stage of the analysis, since the theoretical potential definition is supposed to be as broad. There are 7 different wind speed classes and corresponding wind power density classes as shown in the following table. The wind speed classes are calculated by the author and are basically a power law extrapolation of the wind speed classes at 10 m as suggested by the National Renewable Energy Laboratory (NREL, 2011). Furthermore, the wind power density classes are enumerated via the methodology described in W. Cliff report, which estimates the wind power density as follows based on the Rayleigh distribution (Cliff, 1977) :

$$WPD = \frac{1}{2} \cdot \overline{\rho \cdot V^3} = \frac{1}{2} \cdot \rho \int_0^{\infty} V^3 f(V) dV \cong 0.955 \cdot \rho \cdot \bar{V}^3$$

where ρ is in Kg/m^3 , $\bar{V}$ is the mean wind speed in m/s and $f(V)$ the Rayleigh distribution. This formula is used to define wind power density classes and not to calculate the wind power density at each grid cell, which is a procedure stated in the previous paragraph and accounts for the different densities at each grid cell. The first part accounts the average of the cube of the wind speed while the last part the cube of the average wind speed. Since the wind turbine models chosen are capable of operating in low wind speeds the upper limit of the first wind speed class is set to 4.4m/s.

¹² The air density in this analysis is not assumed constant and equal to 1.225kg/m^3 as in most studies (Shata, 2012), (F. Amar et al, 2007). This figure is valid for air density at sea level at 1atm and 20°C . A more accurate calculation method of the air density is applied, since the elevation plays important role to the variance of it.

¹³ If considering 2 different temperatures ($\Delta T=20^\circ\text{C}$) at the same site, the WPD at the lower temperature would be just 7% higher than the WPD at the higher one. This calculation is rather simple.

The ideal gas formula is used to calculate the air density; $\rho = \frac{P}{RT}$

Table 3.2: Wind power classes' definition (Author's calculations)

Wind Power Class	Wind Speed (m/s)	WPD (W/m²)	Resource Potential
1	0-5.9	0-243	Not suitable
2	5.9-6.9	243-378	Marginal
3	6.9-7.5	378-500	Fair
4	7.5-8.1	500-616	Good
5	8.1-8.6	616-748	Excellent
6	8.6-9.4	748-978	Outstanding
7	>9.4	>978	Superb

The map below demonstrates the wind power potential in terms of wind power density at each grid point based on the wind power classes stated in the table above. This map gives a first insight about the areas with high wind power potential on the continent. The legend showing the wind power density classes as well as coordinate system and scale are stated under the map. The map is obtained by combining analyzed wind speed data from NASA (daily averages), the elevation map of Africa and the African map which shows the counties' boundaries.

Initially it should be stated that there are several countries that represent sufficient wind power potential. The lowest potential is experienced in central Africa, while the highest one along the African coastline and more specifically in Morocco, Western Sahara, Somalia, Kenya and the northeastern part of Madagascar. The darker green zones symbolize higher wind power density, while the beige zones symbolize significantly low wind power density. Despite the fact that beige color signifies low wind power density, there are of course some suitable sites for wind power installations. This cannot be seen in the following map, since the data used at this point of the analysis are not of high resolution, while the assessment area is expressively large to demonstrate the few areas with high potential in the beige zones. Detailed results are obtained in the following paragraphs since a thorough analysis concerning the wind power potential is performed.

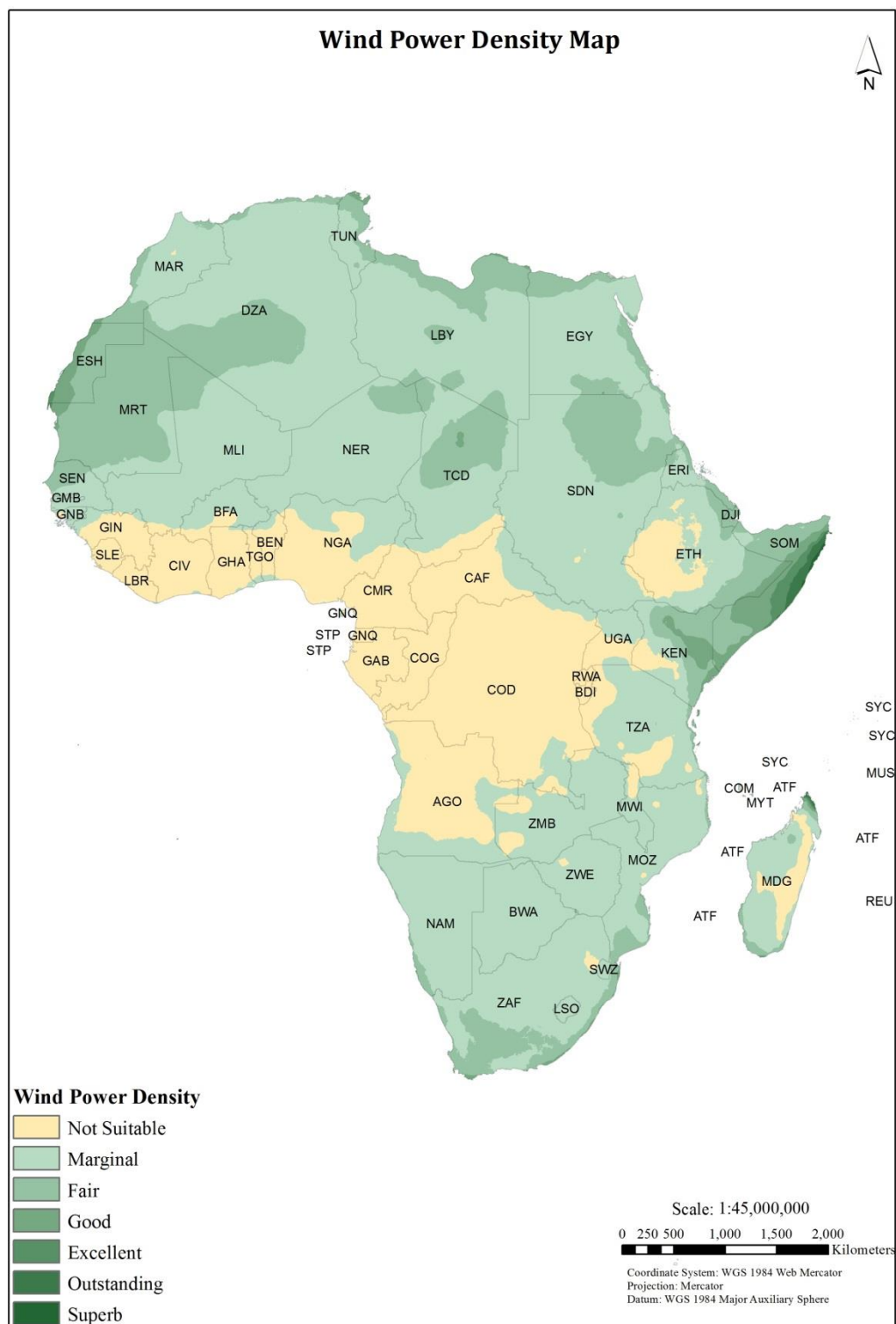


Figure 3.1: Wind power density map of the African continent

3.3 The geographical potential

The first reduction in the theoretical potential in this study is the restriction to onshore areas only. At present, the wind energy industry is showing much interest in offshore wind energy applications. The future of wind energy might be significantly offshore in countries with a sizeable coastal region and land scarcity. The technical potential of offshore wind electricity production is considered to be large and generation costs may decrease to cost-effective levels. However, offshore wind energy is excluded in this study because insufficient wind speed data are available to justify a proper analysis of the offshore wind energy potential in Africa.

The onshore area available for wind power is further restricted to areas that are suitable for wind turbine installation. The exclusion parameters that are considered are the following:

- *Water bodies*: any type of water bodies, lakes, sea, rivers, wetlands, floodplains and saline pans are to be protected to preserve the natural wealth (Serwan M.J.Baban et al, 2000).
- *Protected areas*: the natural wealth of the continent should be shielded and the biodiversity values should be sustained
- *Forest areas*: forestry increases turbulence and wind shear; hence, the turbine loading increases and the design conditions might be exceeded. This also may effect operating and maintenance costs over the project lifetime. Also, the forestry reduces the wind speeds above canopy, which leads to reduced energy production; hence, reduced income (Andrew Tindal, 2008).
- *High elevation areas*: Sites with higher elevation than 2000m are excluded from this analysis, due to the fact that high elevation results in high transportation costs, high transmission costs and lower air density (Baban Serwan, 2000).
- *Urban built-up areas*: highly urbanised or otherwise densely populated regions are severely constrained as is evident in potential assessment and planning studies at national level (M.Hoogwijk et al, 2004). Also, due to the noise and vibration generated from wind turbines it is important to ensure that wind farms are located outside the residential area (Al-Yahyai et al, 2012).
- *Distance to urban areas*: a further criterion sets a maximum distance of 200km to the nearest city (50,000 inhabitants or more). This avoids accounting for extremely rural areas (e.g. associated with prohibitively high transmission costs for large scale renewable electricity systems).
- *Sloped areas*: areas with slopes larger than 10 degrees are excluded from investigation. Steeper slopes give rise to stronger wind flow, but on the lee of ridges steeper slopes give rise to high turbulence (Al-Yahyai et al, 2012; Baban Serwan, 2000).

- *Agricultural zones:* Areas with agricultural activities are excluded in order to sustain biodiversity. Large scale wind farms are not intended to be built in those areas.
- *Distance to existing grid lines:* An alternative, optional criterion can be established to exclude all areas exceeding a certain distance to the existing electricity grid. This can be used to avoid areas with prohibitively high transmission costs.

After applying the above geographical constraints, the total amount of land area available for wind turbine installation is obtained and expressed in km². The exclusion map is being composed by using the GIS overlay function to combine all the layers (i.e. all the exclusion criteria).

3.4 GIS-based approach

In order to modify and treat the exclusion areas, the map data related to the above criteria (geospatial data) should be imported in a Geographic Information System (GIS). The same applies for the results obtained from wind energy assessment. GIS is a tool capable of storing location dependent features and characteristics in an elegant way making it possible to combine certain features of a location and tabulate areas (ESRI, 2009). GIS is a powerful and quick tool for analysing geographic data and obtaining valuable information from the results. These geospatial data used in this analysis are publically available. GIS provides with the possibility to interlink several features and characteristics of an area and facilitates the combination of the different selection criteria aiming to the creation of an ultimate exclusion map.

To illustrate, natural geographic features (such as slopes, elevation, land use, water bodies) can also be combined with available social data (e.g. population density or poverty data) or technical infrastructure (such as roads or power grids). A complete list with the publically available GIS data used in this report plus additional resources are stated in the previous chapter. The restriction maps are constructed and reclassified in such a way that shows the suitable and not suitable areas for wind farms. All the maps have a grid size 1km×1km and this allows implementing algebraic calculations with all the maps. An ultimate restriction map is being produced by overlaying all the restriction maps.

Below all the restriction zones are illustrated, i.e. 8 different maps which show how each restriction affects the wind power potential.

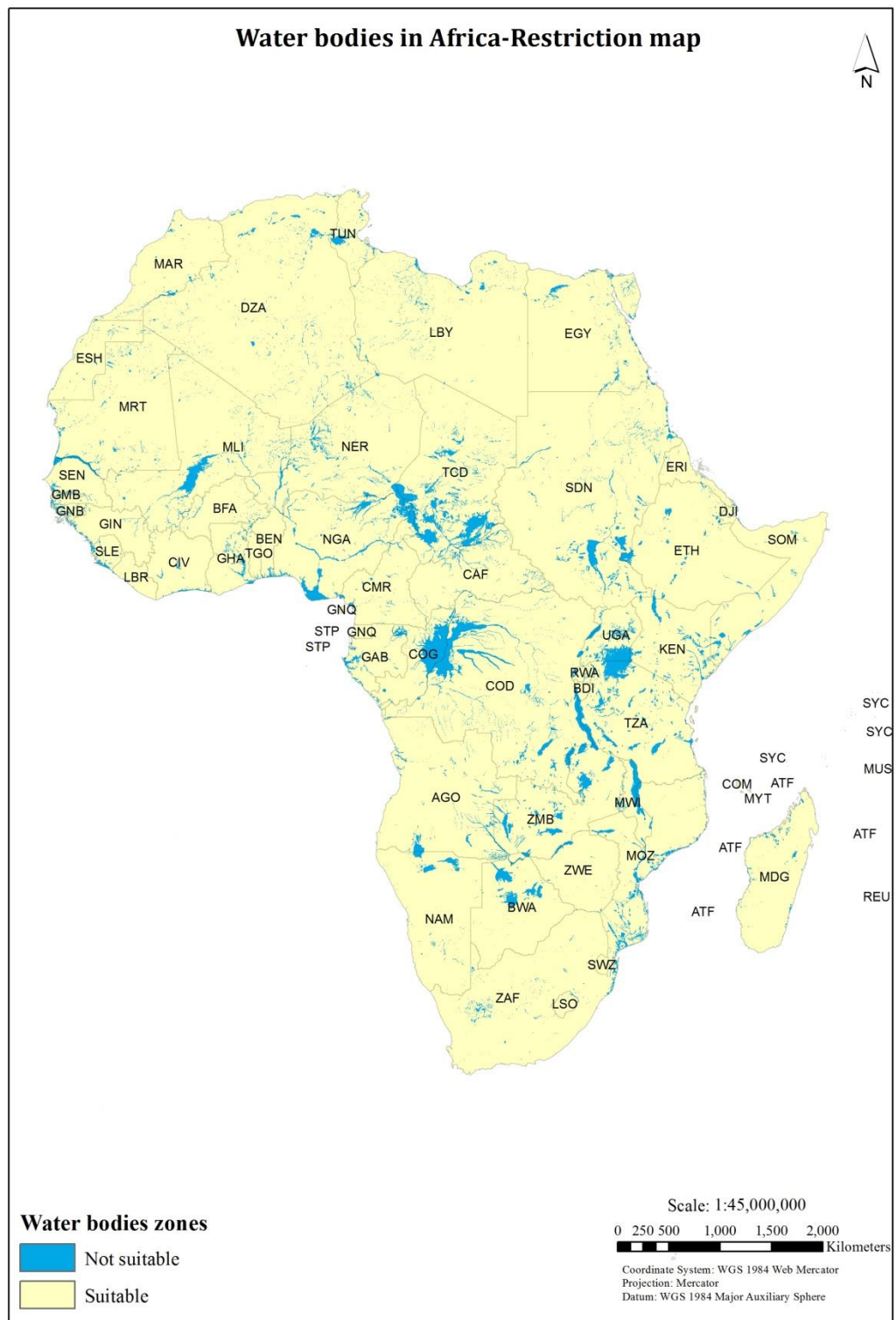


Figure 3.2: Restriction map for water bodies in Africa

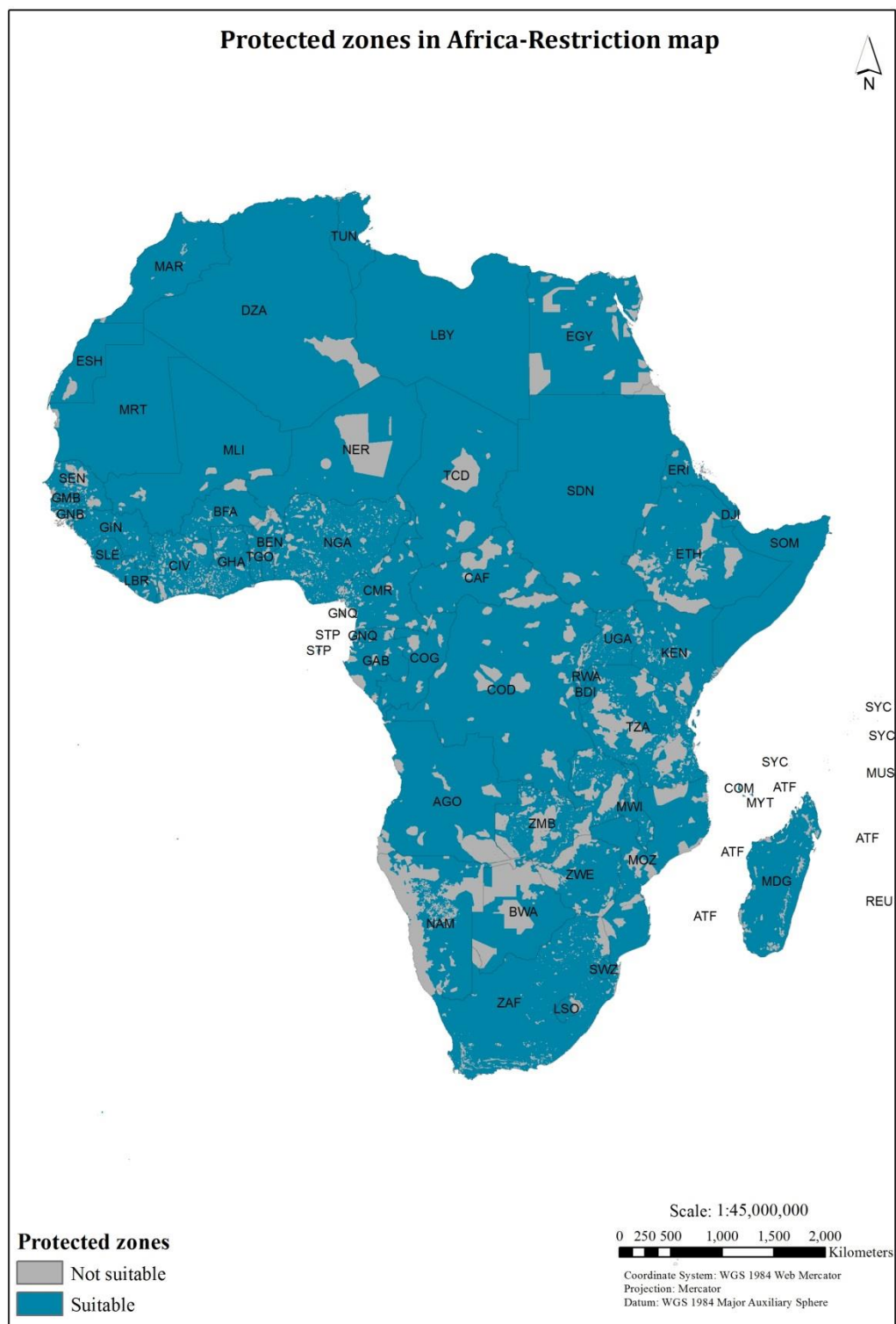


Figure 3.3: Restriction map for protected areas in Africa

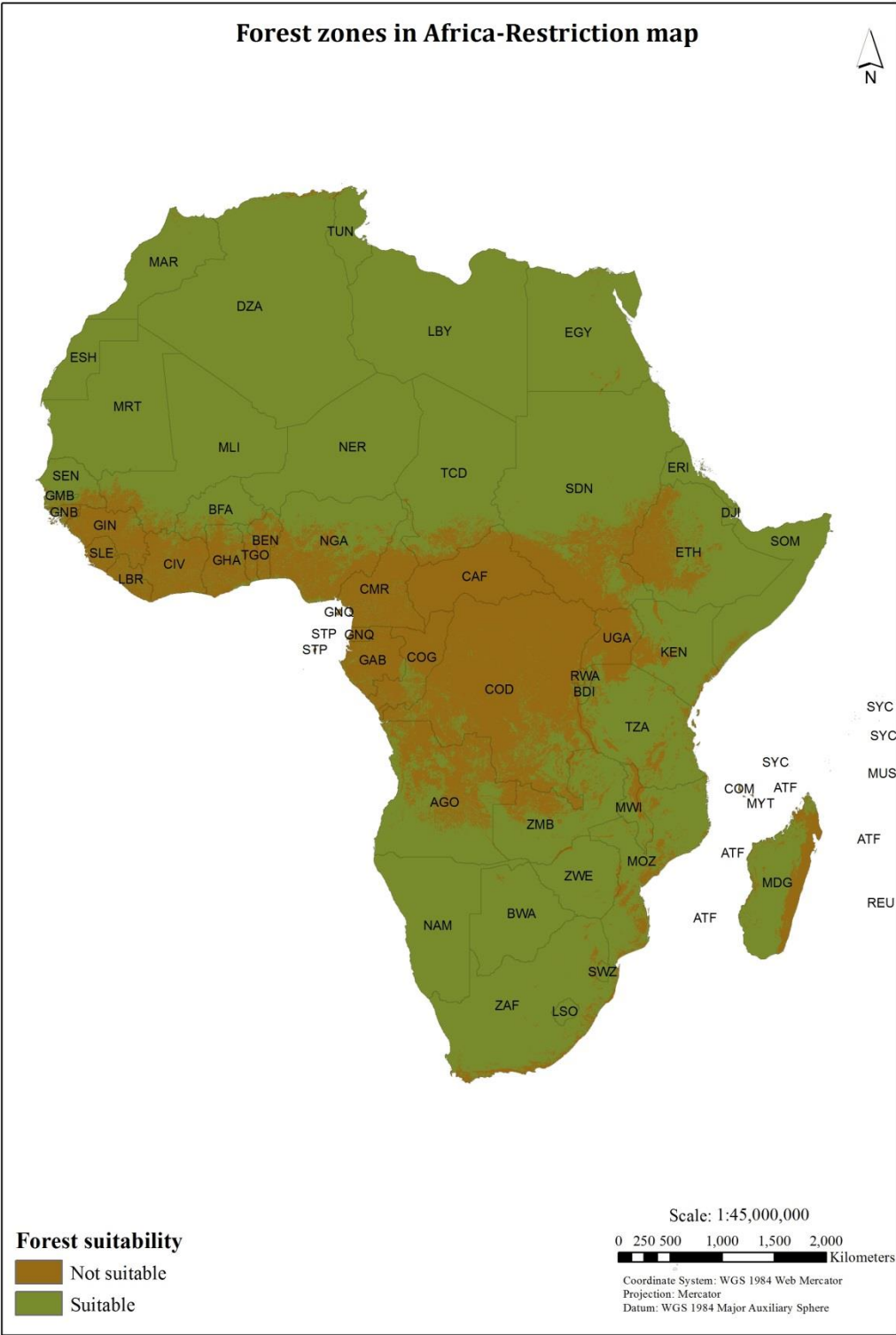


Figure 3.4: Restriction map for forest zones with higher than 20% tree covered areas in Africa

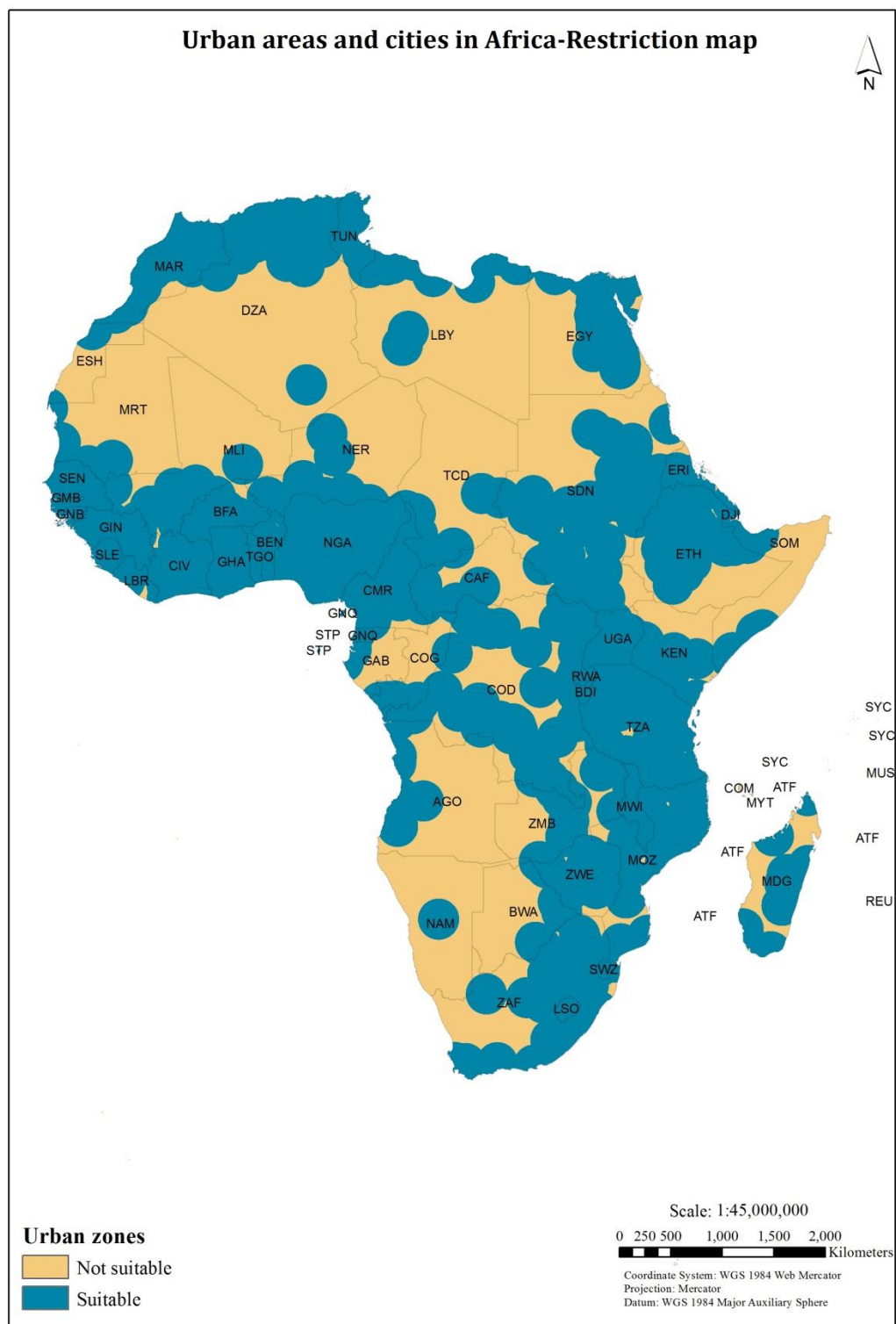


Figure 3.5: Restriction for extremely rural areas in Africa



Figure 3.6: Restriction map for areas with higher elevation than 2000m



Figure 3.7: Restriction map for slopes higher than 10° in Africa

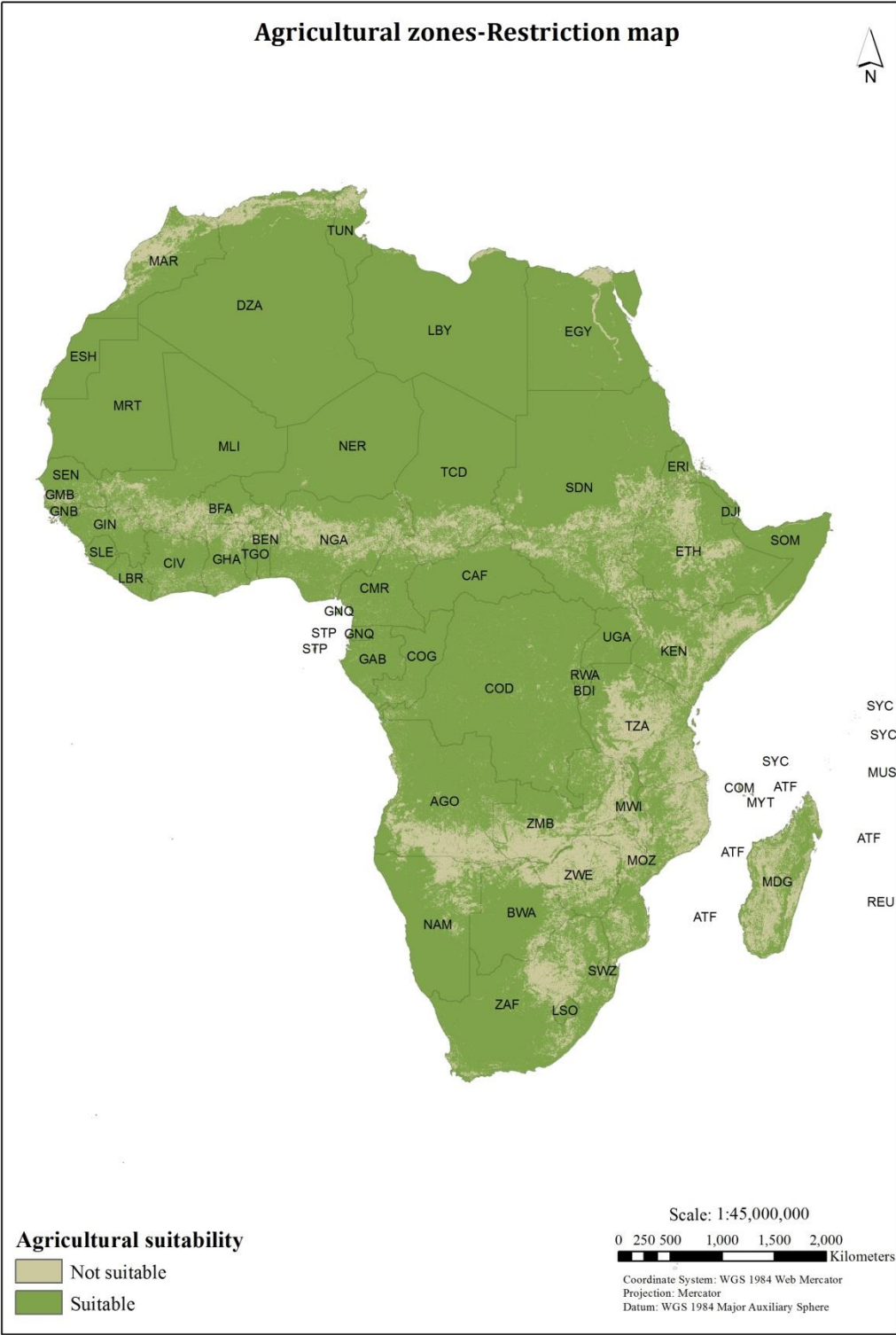


Figure 3.8: Restriction map for all the agricultural zones in Africa

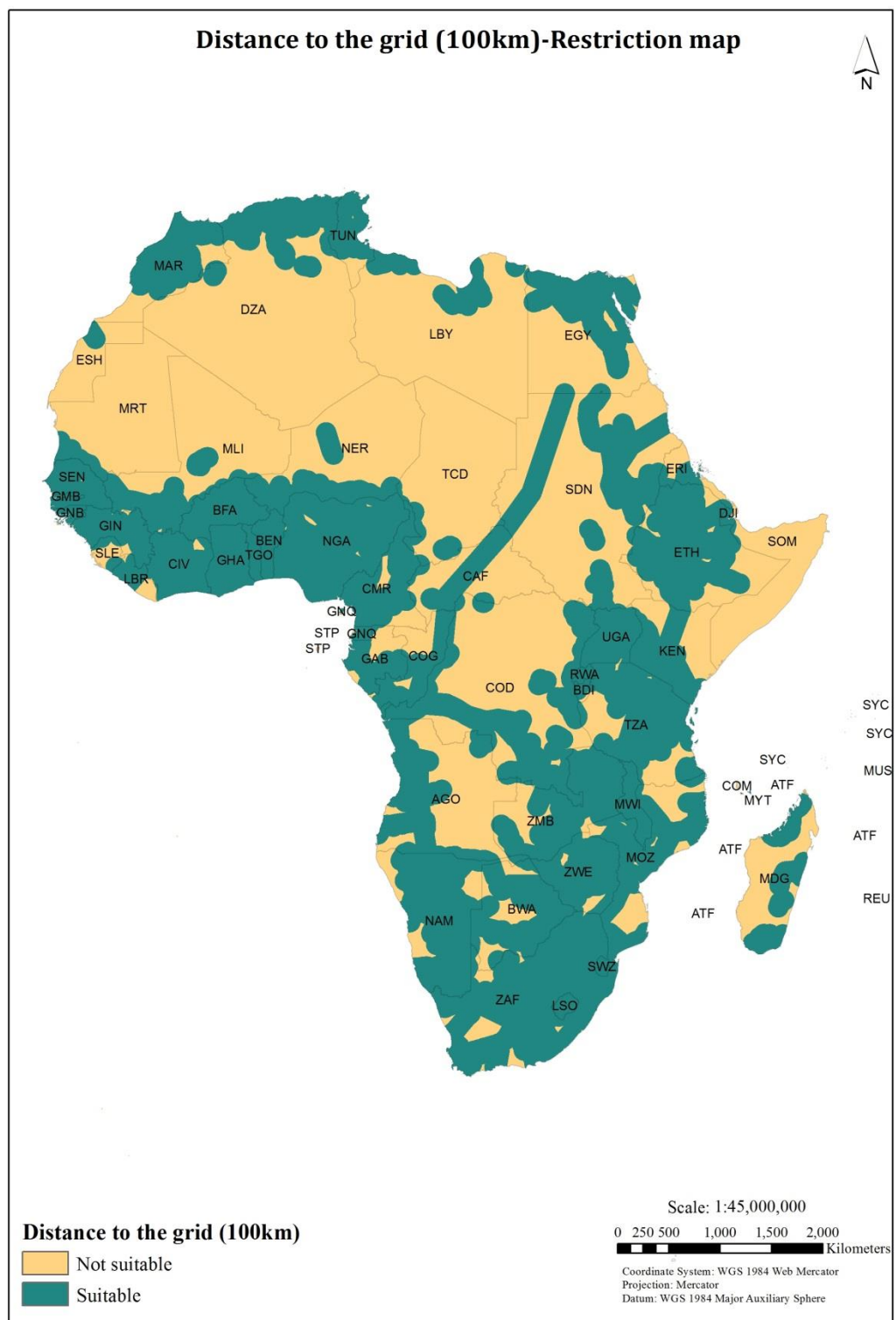


Figure 3.9: Restriction for areas exceeding a 100km distance to the electrical grid

By combining all the previously stated restriction maps, an ultimate suitability map is constructed using a GIS raster calculator and a spatial analyst tool for extracting specific areas from the African map. The coordinate system used through this approach is projected Web Mercator and the geodetic system WGS 1984. Wherever geographic coordinate system was given, it was converted to the projected one using a data management tool able to transform a raster dataset from one projection to another in order to diminish the error in less than half pixel.

In fact there are two suitability maps obtained from the analysis and presented below. The first one includes the optional criterion which regards the distance to the grid to be less than 100km, while the second one not. These ultimate suitability maps are used at a later stage in order to estimate the technical wind power potential. The zones with light blue represent the appropriate areas for setting up wind farms concerning the geographical constraints. The light gray areas are not suitable and thus excluded from the following analysis. The geographical wind power potential is synopsized in the table 8-3. Burkina Faso, South Africa, Djibouti, Cape Verde, Tunisia, Gambia and Senegal indicate a high area availability percentage, while Chad, Democratic Republic of Congo, Republic of Congo, Central African Republic, Liberia and a few other smaller countries point toward lower area availability. More detailed information about which suitability factors affect the most the geographical potential of each country is provided in the Appendix.

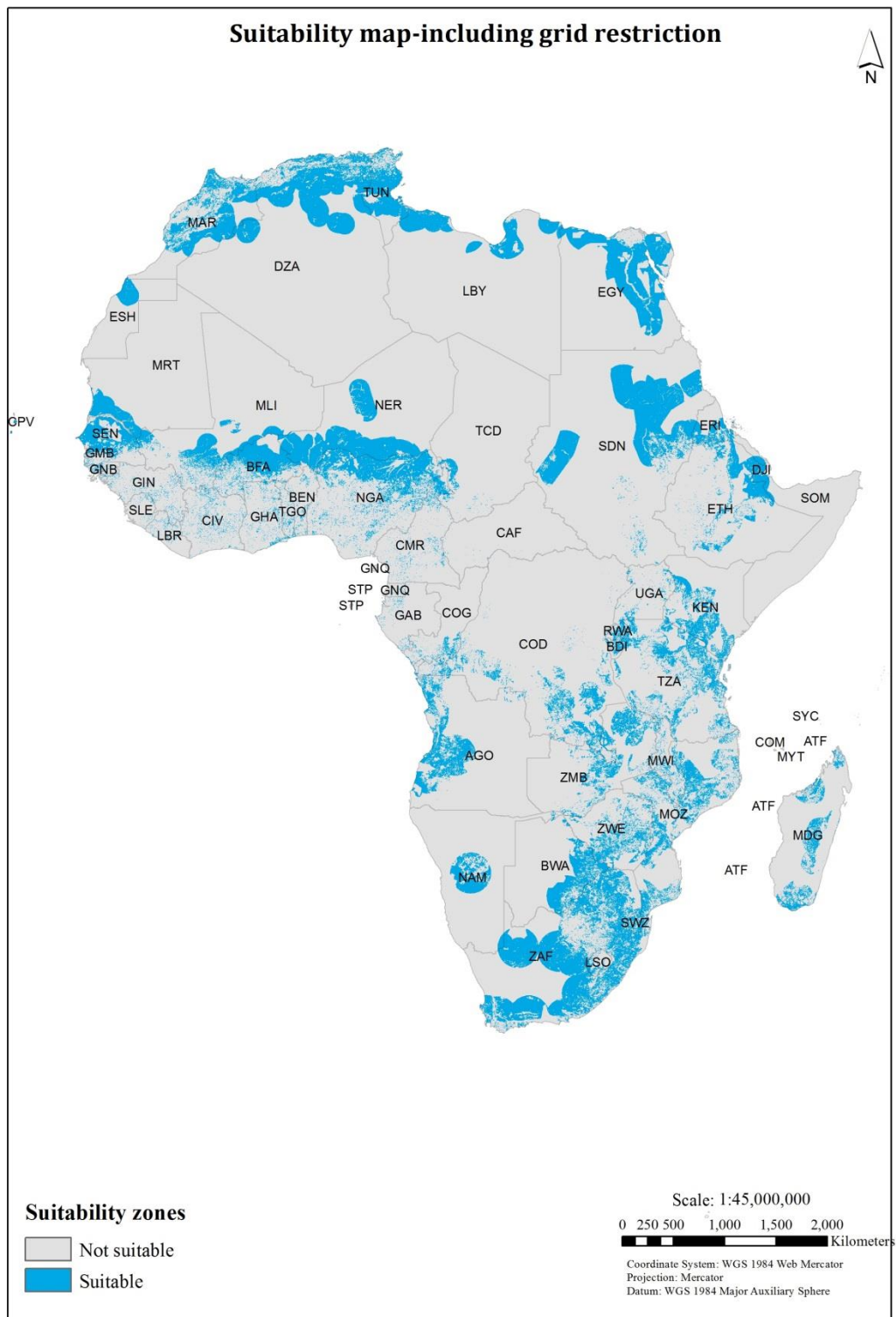


Figure 3.10: Suitability map for wind power installations in Africa-including grid restriction

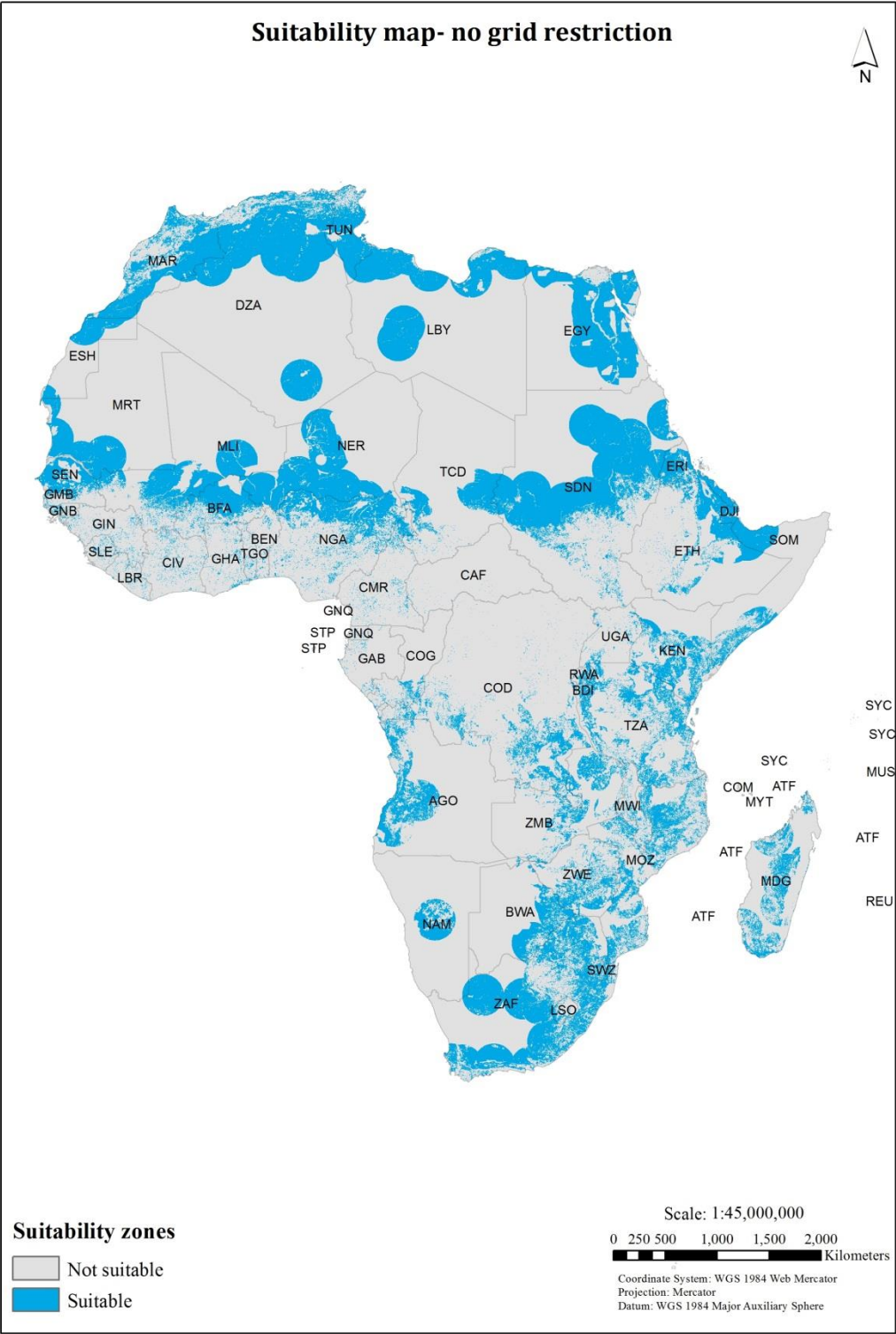


Figure 3.11: Suitability map for wind power installations in Africa-no grid restriction

Table 3.3: Geographical wind power potential

COUNTRY	Total area PCS ¹⁴ (km ²)	Grid restriction		No grid restriction	
		Total Available Area for Wind Farms (km ²)	Percentage of area availability	Total Available Area for Wind Farms (km ²)	Percentage of area availability
Algeria	3012792	438271	14.55%	816276	27.09%
Angola	1318689	165387	12.54%	188331	14.28%
Benin	119543	8379	7.01%	8381	7.01%
Botswana	678819	128208	18.89%	128889	18.99%
Burkina Faso	288184	134644	46.72%	134644	46.72%
Burundi	27235	10047	36.89%	11941	43.84%
Cameroon	474587	35458	7.47%	37077	7.81%
Cape Verde	4082	2180	53.41%	2232	54.68%
Central African Republic	633081	1599	0.25%	3009	0.48%
Chad	1380620	21002	1.52%	159129	11.53%
Comoros	1613	0	0.00%	0	0.00%
Côte d'Ivoire	329710	25354	7.69%	25498	7.73%
Democratic Republic of the Congo	2363172	149941	6.34%	193925	8.21%
Djibouti	22648	16898	74.61%	20193	89.16%
Egypt	1234950	355348	28.77%	512080	41.47%
Equatorial Guinea	27143	1277	4.70%	1330	4.90%
Eritrea	131483	45101	34.30%	93414	71.05%
Ethiopia	1163504	137791	11.84%	173867	14.94%
French Southern Territories	12	0	0.00%	0	0.00%
Gabon	266322	5522	2.07%	5901	2.22%
Gambia	11362	5296	46.61%	5296	46.61%
Ghana	244728	20674	8.45%	20674	8.45%
Guinea	254518	16322	6.41%	17244	6.78%
Guinea-Bissau	34707	5368	15.47%	5368	15.47%
Kenya	586779	106230	18.10%	149855	25.54%
Lesotho	40422	9811	24.27%	9811	24.27%
Liberia	97684	5711	5.85%	7907	8.09%
Libya	2052769	181004	8.82%	526781	25.66%
Madagascar	668296	89639	13.41%	146808	21.97%
Malawi	125480	17546	13.98%	17546	13.98%
Mali	1386749	135788	9.79%	308017	22.21%
Mauritania	1192094	75724	6.35%	232921	19.54%
Mayotte	378	0	0.00%	0	0.00%

¹⁴ Projected coordinate system; WGS 1984 Web Mercator

Morocco	565088	214994	38.05%	333347	58.99%
Mozambique	869602	201474	23.17%	254470	29.26%
Namibia	968078	88576	9.15%	89111	9.20%
Niger	1310816	332100	25.34%	480322	36.64%
Nigeria	941177	283661	30.14%	284480	30.23%
Republic of Congo	344424	18136	5.27%	20948	6.08%
Rwanda	25400	9405	37.03%	9405	37.03%
Sao Tome and Principe	946	0	0.00%	102	10.78%
Senegal	210459	97888	46.51%	97888	46.51%
Seychelles	43	0	0.00%	0	0.00%
Sierra Leone	74245	2829	3.81%	7080	9.54%
Somalia	645680	7476	1.16%	139238	21.56%
South Africa	1604675	706162	44.01%	732865	45.67%
Sudan	2693029	520733	19.34%	1028530	38.19%
Swaziland	21702	12185	56.15%	12185	56.15%
Tanzania	960923	174057	18.11%	220352	22.93%
Togo	58670	6850	11.68%	6850	11.68%
Tunisia	226346	133441	58.95%	164865	72.84%
Uganda	243140	14686	6.04%	14686	6.04%
Western Sahara	325100	36891	11.35%	90476	27.83%
Zambia	800530	123623	15.44%	125092	15.63%
Zimbabwe	439690	92063	20.94%	101568	23.10%

4 Wind Energy Assessment

In this section a large quantity of wind data has been collected in order to estimate the energy production from potential wind farms. The power available in the wind and the power curve of a chosen wind turbine can indicate the energy production of a wind turbine at a certain site. In the following paragraphs it is explained how the wind speed data are analyzed and handled in order to obtain the potential energy production.

4.1 Area of interest

The wind resource assessment concerns the whole continent. So the analysis considers the geographical area that is formed by a rectangle with its four corners being assigned at the most northerly, the most southerly, the westernmost and the most easterly points. These 4 corner coordinates are the following ones (Merriam-Webster, 2001):

- Northernmost point: Ras ben Sakka in Tunisia (37°21' N)
- Southernmost point, Cape Agulhas in South Africa (34°51'15" S)
- Westernmost point, Pointe des Almadies in Cap Vert in Senegal, (17°33'22" W)
- Easternmost point, Ras Hafun in Somalia (51°27'52" E)

4.2 Wind Speed Adjustment

The wind speed changes with altitude because of frictional effects at the surface of the earth. Since the daily wind speed data are given at 10m, a proper scaling should be applied in order to obtain wind speed figures at the hub height of a chosen wind turbine model, i.e 80m. The scaling can be carried out by implementing the power law, which is used by many wind energy researchers in order to extrapolate the reference wind speed to the hub height (J. Rogers et al, 2009). It represents a simple model for the vertical wind speed profile and its basic form is:

$$\frac{U(z)}{U(z_r)} = \left(\frac{z}{z_r}\right)^\alpha$$

where $U(z)$ is the wind speed at height z , $U(z_r)$ is the reference wind speed at height z_r , and α is the power law exponent also known as wind shear exponent; one way of handling α is from the reference values, which are already known. The expression form is:

$$\alpha = \frac{0.37 - 0.088 \ln(U_r)}{1 - 0.088 \ln\left(\frac{z_r}{10}\right)}$$

where U is given in m/s and z_r in m.

4.3 Probability distribution

The wind speed distribution describes the variation of the wind with respect to time and the effect of the varying wind on the power output of the turbine (Z. Olaofe et al, 2012). Two probability distributions are commonly used in wind data analysis the Rayleigh and the Weibull. The Rayleigh distribution uses one parameter: the mean wind speed, while the Weibull distribution is based on two parameters and, thus, can better represent a wider variety of wind regimes and therefore is frequently used to characterize a site.

As above stated, the usage of the Weibull probability density function requires the knowledge of two parameters: k , a shape factor, which is a measurement of the width of the distribution, and A , a scale factor, which is closely related to the mean wind speed. Both of these parameters are functions of mean wind speed and standard deviation of the measured data (J. Rogers et al, 2009). The Weibull distribution is expressed with the following probability density function $f(U)$:

$$f(U) = \frac{k}{A} \cdot \left(\frac{U}{A}\right)^{k-1} \cdot \exp\left(-\left(\frac{U}{A}\right)^k\right),$$

where U is the wind speed (ranging from 0 to 25 m/s) and the factors k and A can be calculated from the next formulas:

$$k = \left(\frac{\sigma_U}{U_{mean}}\right)^{-1.086}, \quad A = \frac{U_{mean}}{\Gamma(1+\frac{1}{k})}$$

where σ_U the standard deviation of the individual wind speed averages and U_{mean} the mean wind speed in m/s

$$U_{mean} = \frac{1}{n} \sum_{i=1}^n U_i$$

$$\sigma_U = \left[\frac{1}{n-1} \sum_{i=1}^n (U_i - U_{mean})^2 \right]^{0.5}$$

and Gamma function is given by the next equation:

$$\Gamma(x) = \int_0^{\infty} e^{-t} \cdot t^{x-1} dt$$

The Rayleigh distribution is used in cases where the wind speed data are limited and only the mean wind speed is available. This is the simplest velocity probability distribution to represent the wind resource. The probability density function is given by:

$$f(U) = \frac{\pi}{2} \cdot \frac{U}{U_{mean}^2} \cdot \exp\left(-\frac{\pi}{4} \left(\frac{U}{U_{mean}}\right)^2\right)$$

In this analysis the Weibull distribution is performed but its results are compared with the Rayleigh's in order to illustrate the importance of choosing this method.

4.4 Distribution adjustment

The wind speed data provided from Vortex are of the highest resolution compared to all the other wind speed data. This infers that these data will be the basis of the analysis, while the data from NASA are used to obtain a more precise wind speed distribution. Vortex provided annual data with grid cell size about $0.067^\circ \times 0.067^\circ$ while the data from NASA Climatology Resource for Agro-climatology are of a coarser resolution but on a daily basis; $1^\circ \times 1^\circ$.

The wind speed distribution that is obtained as described in the previous paragraph (for the NASA daily data), should now be applied to the higher resolution grid cells. This can be realized by implementing the method of the bilinear interpolation.

It can be perceived as an extension of linear interpolation for interpolating functions of two variables on a regular 2D grid; in this analysis the two variables are latitude (Y) and longitude (X). This function represents the standard deviation. The reason why standard deviation is the parameter that needs to be calculated at all grid points is due to the fact that it is the value missing to enumerate the Weibull scale and shape parameters.

Suppose that the standard deviation at a certain point $S(X,Y)$ is unknown. Also, suppose that the standard deviations at two neighbor points ($Long_1, Lat_1$) and ($Long_2, Lat_2$) are known. By applying two times linear interpolation, first in the x-direction and then in the y-direction, the required standard deviation at $S(X,Y)$ is obtained. Suppose that in the following sketch, the $S(X,Y)$ at a point (X,Y) while the standard deviation at the neighboring points are known, see Figure 4.1.

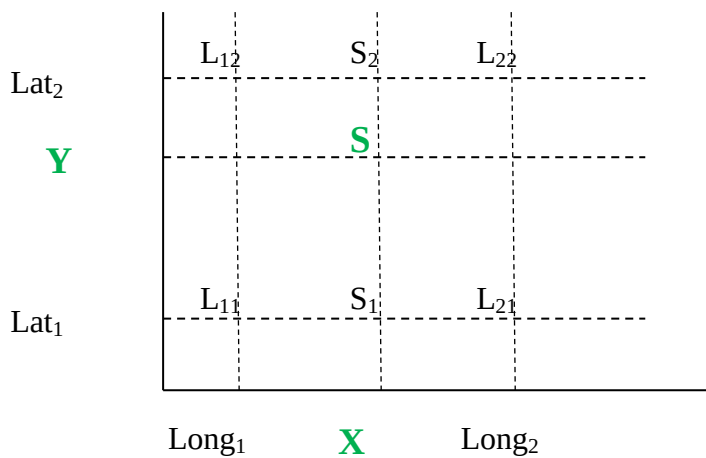


Figure 4.1: Schematic representation of the bilinear interpolation

The algorithm that demonstrates the above is:

$$f(S_1) \approx \frac{Long_2 - X}{Long_2 - Long_1} \cdot f(L_{11}) + \frac{X - Long_1}{Long_2 - Long_1} \cdot f(L_{21})$$

where $S_1 = (X, Lat_1)$

$$f(S_2) \approx \frac{Long_2 - X}{Long_2 - Long_1} \cdot f(L_{12}) + \frac{X - Long_1}{Long_2 - Long_1} \cdot f(L_{22})$$

where $S_2 = (X, Lat_2)$

The interpolation in the y-direction follows (latitude axis).

$$f(S) \approx \frac{Lat_2 - Y}{Lat_2 - Lat_1} \cdot f(S_1) + \frac{Y - Lat_1}{Lat_2 - Lat_1} \cdot f(S_2)$$

Ultimately the required standard deviation for the point (X,Y) is expressed by applying the above equations:

$$f(S) \approx \frac{Lat_2 - Y}{Lat_2 - Lat_1} \cdot \left[\frac{Long_2 - X}{Long_2 - Long_1} \cdot f(L_{11}) + \frac{X - Long_1}{Long_2 - Long_1} \cdot f(L_{21}) \right] + \frac{Y - Lat_1}{Lat_2 - Lat_1} \cdot \left[\frac{Long_2 - X}{Long_2 - Long_1} \cdot f(L_{12}) + \frac{X - Long_1}{Long_2 - Long_1} \cdot f(L_{22}) \right]$$

The above after proper factorization would give the subsequent bilinear interpolation equation:

$$f(S) \approx \frac{1}{Lat_2 - Lat_1} \cdot \frac{1}{Long_2 - Long_1} \cdot [f(L_{11})(Long_2 - X) \cdot (Lat_2 - Y) + f(L_{21})(X - Long_1) \cdot (Lat_2 - Y) + f(L_{12})(Long_2 - X) \cdot (Y - Lat_1) + f(L_{22})(X - Long_1) \cdot (Y - Lat_1)]$$

Bilinear interpolation uses the value of the four nearest input cell centers to determine the value on the output raster (ESRI, 2009). The new value for the output cell is a weighted average of these four values, adjusted to account for their distance from the center of the output cell. This interpolation method results in a smoother-looking surface than can be obtained using nearest neighbor. Bilinear interpolation is preferred for data where the location from a known point or phenomenon determines the value assigned to the cell, such as elevation and slope. Such phenomena represented as continuous surfaces are most appropriately resampled using bilinear interpolation.

This method is widely used to sample and resize image data. It is basically using the neighbor values to get the optimal value for the required point (Jian Jing Liu, 2010). By doing so, the wind speed distribution of a coarse resolution can be utilized and assigned in a fine way to grid cells of higher resolution. Of course such method is not

flawless since it does not provide the perfect distribution but the optimal one based on the available data. It can provide though an overview of how the wind should be distributed; otherwise it should have been an only wind mean based distribution.

4.5 Machine Power

In this analysis, there are 2 different wind turbine models considered and evaluated. The nominal capacity of these wind turbines is 2 MW. There are no larger machines selected for evaluation since larger size would mean barriers in their transportation through some parts of the African continent. Also, larger than 2 MW turbines would cause more clear visual disturbance and African inhabitants might oppose to such large machines. The chosen wind turbines for evaluation are the Vestas V90 (swept area: 6362m^2 , rotor diameter: 90m) and the Enercon E82 (swept area: 5281m^2 , rotor diameter: 82m). These particular turbines are selected due to the fact that they both perform well in low and middle wind speeds. Further characteristics of these 2 wind turbines are provided in the Appendix (Tables A-1 and A-2).

The power output of the wind turbines varies with wind speed and every wind turbine has a characteristic power performance curve. This curve is advantageous since it facilitates the prediction of the energy production of a wind turbine without considering the technical details of its various components. The power curve gives the electrical power output as a function of the hub height wind speed and illustrates three important characteristic velocities:

- **Cut-in velocity:** the minimum wind speed at which the machine will deliver useful power
- **Rated velocity:** the wind speed at which the rated power (generally the maximum power output of the electrical generator) is reached
- **Cut-out velocity:** the maximum wind velocity at which the turbine is allowed to deliver power (usually limited by engineering design and safety constraints).

In the following figures, the power curves of these two models are presented with black color while with dark blue it is sketched the power coefficient of each turbine. The power coefficient is defined as the ratio of the rotor power to the available power in the wind.

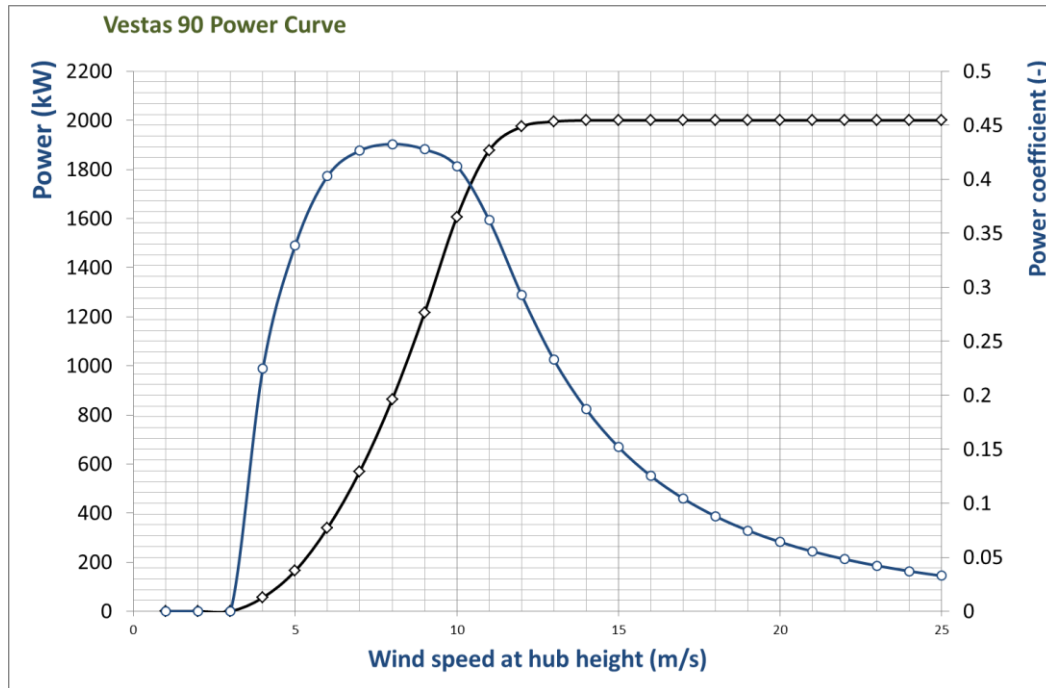


Figure 4.2: Vestas V90 Power Curve (RETScreen, 2013)

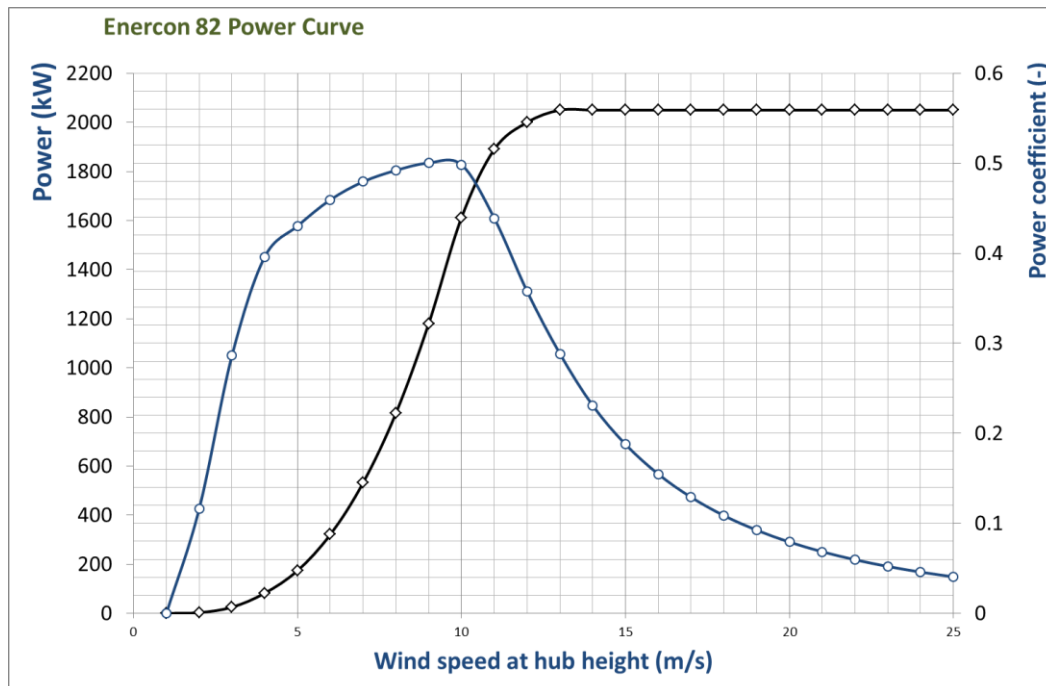


Figure 4.3: Enercon E82 Power Curve (Enercon, 2013), (RETScreen, 2013)

The above presented power curves together with the wind speed distribution which is provided at each grid point are used in order to calculate the energy production at all the given grid points on the whole African continent. A grid point in this analysis is defined as a point in the African map that is defined by two coordinates (latitude and longitude). The grid size varies from $0.067^{\circ} \times 0.067^{\circ}$ (Vortex data) to $1^{\circ} \times 1^{\circ}$ (NASA data). It is worth providing an example on how to estimate the energy production at a certain point. The same methodology is applied to all the grid points of the whole continent.

Let the grid point have the following coordinates (17.972°E, 34.941°S): The mean wind speed is 6.942 m/s based on 366 measurements in 2012. The standard deviation of the measured data is 3.251 m/s. By applying the Weibull distribution formulas, the shape factor k and the scale factor A are calculated; $k=2.280$, $A=7.837$ and hence the Weibull distribution is obtained. In the next graph, both the Weibull distribution (blue line) and the power curve of the Vestas wind turbine (black line) are shown. It should be noted that this procedure is performed for all the coordinates with measured wind speed data and for both wind turbine models.

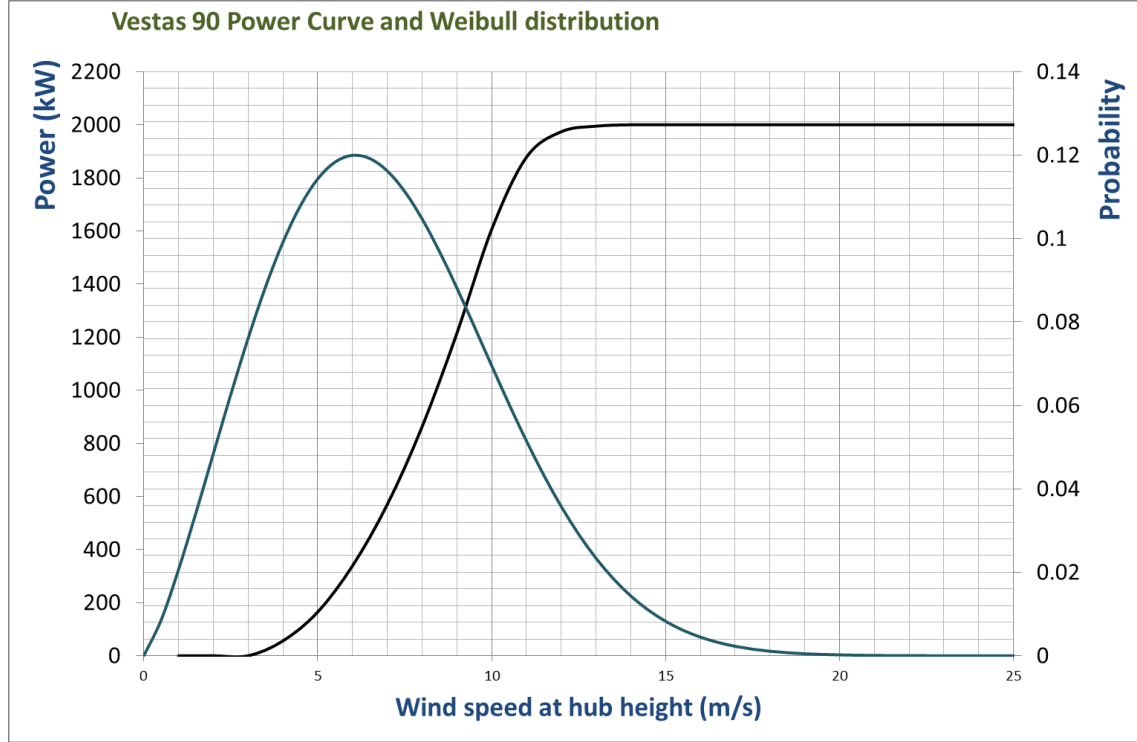


Figure 4.4: Power curve and Weibull distribution at a certain grid point

The expected power production by the Vestas wind turbine at this particular point at an annual basis is given by the intersection of the two figures at the following graph and the algebraic formula of this intersection is the subsequent:

$$P_{expected} = \int_{U=0}^{U=cut_{out}} P(U) \cdot f(U) dU \approx \sum_{wind\ class} P(U) \cdot f(U)$$

where $P(U)$ is the power curve and $f(U)$ the probability of occurrence of wind class U . The wind classes considered for this Weibull approach range from 0 to 25 m/s and each class' range is 0.5 m/s. The expected yearly energy yield can be expressed with the following formula:

$$E_{expected} = \mu \cdot T \int_{U=0}^{U=cut_{out}} P(U) \cdot f(U) dU \approx \mu \cdot T \sum_{wind\ class} P(U) \cdot f(U)$$

where μ is the availability factor of the wind farm, and T is the time in hours of a full year.

The availability factor of the wind farms “ μ ” is typically between 94% and 97% while recent data about modern wind farms indicate that availability is now on the order of 97–99% (J. Rogers et al, 2009). Availability factor is the percentage of time that a wind turbine is available (or capable) to produce electricity. Hypothetically a wind turbine could be available 100% of the time, however you must factor in planned outages for scheduled maintenance and inspections which could eat up 2 to 3% of the available operating time, and forced outages such as breakdowns, grid outages, and severe weather conditions that could also take up an additional 2 to 3% of the available operating time.

After applying the above parameters to the above formula, the expected energy yield is obtained:

$$E_{expected} \approx 0.97 \cdot 365.25 \cdot 24 \sum_{wind\ class} P(U) \cdot f(U) \approx 7732 MWh$$

The expected energy yield and the nominal power of the wind turbine are used to calculate the capacity factor of a wind turbine at the map point that is being evaluated. The capacity factor of a wind turbine at a given site is defined as the ratio of the energy actually produced by the turbine to the energy that could have been produced if the machine ran at its rated power over a given time period (J. Rogers et al, 2009). It is used as a performance parameter and it is a sufficient indicator for comparing different sites. The capacity factor is given by the next formula:

$$CF = \frac{\text{Actual Energy Production}}{\text{Hypothetical Energy Production at Rated Power}} = \frac{E_{expected}}{P_{rated} \cdot T}$$

At this case the capacity factor is calculated as follows:

$$CF = \frac{7732}{2 \cdot 365.25 \cdot 24} = 43\%$$

This is a rather high capacity factor and it implies that a wind farm should be installed at this site if the siting criteria are fulfilled. After assessing the wind energy potential with both wind turbines, it is concluded that the Vestas model produces about 28% more than the Enercon one on the whole continent. This is due to the higher power coefficient that Vestas turbine achieves in lower wind speeds and the fact that the average wind speed on the continent ranges in these wind speed levels. It should also be concluded that the difference between Rayleigh and Weibull distribution along the whole continent is slightly more than 5%. However the variation at certain sites exceeds 100% since the shape factor ($k=2$ for Rayleigh) is not valid at those locations. Rayleigh distribution works for calculating the wind power potential on a continental level, but not on a country level. Hence, Vestas model and Weibull distribution are selected for the wind energy assessment.

4.6 Wind Turbines Spacing

In wind farms the wake effect plays significant role to the energy production. The extraction of energy by the wind turbines that are upwind of other turbines result in lower wind speeds at the downwind turbines and increased turbulence. The wake effect can decrease energy production and increase wake-induced fatigue in turbines downwind of other machines. Thus, the wind turbine spacing is a very critical issue when it comes to the siting of a wind farm. It can be concluded that the longer the distance between the turbines, the less wake effect is experienced. However, this would increase the grid connection cost. So a compromise between grid connection costs and wake effect should be done.

Recent studies have shown that for turbines that are spaced 10 rotor diameters, D , apart in the prevailing downwind direction and 5 rotor diameters apart in the crosswind direction, array losses are typically very low (less than 10%) (J. Rogers et al, 2009). These spacing parameters indicate a spacing factor of about 5 MW per km²¹⁵ if considering a rotor diameter $D=90\text{m}$. The spacing rule is represented as a sketch in the figure below. The compass shown in the upper left corner is used to show the prevailing wind speed direction, which in the following case would be about 45°NE. This is a random case, but it is used to illustrate how the spacing rule is obtained.

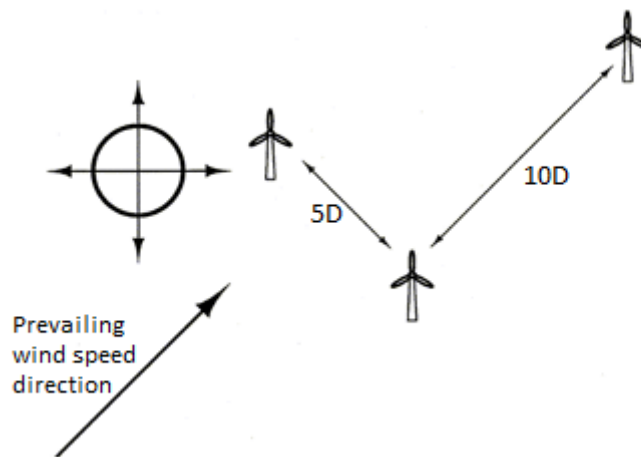


Figure 4.5: Schematic representation of the spacing rule (Author's sketch)

The array losses are also taken into account since they play a significant role in the annual energy yield of a wind farm. Thus the array efficiency is imported into the GIS analysis. It depends on the spacing between turbines and the nature of the wind regime as well as the number of turbines, the size of the wind farm and the frequency distribution of the wind direction (Schallenberg-Rodriguez, 2012). It can mathematically be expressed as:

$$\eta_{ar} = \frac{\text{Annual energy whole array}}{\text{Annual energy if the wind turbines are isolated}}$$

¹⁵ The actual spacing factor obtained from the calculation is 4.94 MW per km². The rounding up to 5 MW per km² is made in order to have a realistic value of how many turbines can be pragmatically installed at a certain km².

The proposed geometry can contribute to the array losses reduction. For this study, it is assumed that the distance between the wind turbines downwind is 10 rotor diameters. This implies array efficiency of about 90% and this is applied in all the wind farms across the continent. Of course a more detailed siting study at a further stage would provide with various array efficiencies. This would require studying in detail the wake effect using tools such as numerical models, artificial neural networks and lately genetic algorithms. But this is not in the scope of this work, so a general fair assumption is made regarding the array losses.

The amount of installed wind power capacity that can be placed in a given area is also influenced by the terrain slope. In fact the density of turbines is generally decreasing as the slope angle increases. As previously stated the slope percentage that is suitable for wind energy production is set to 10°. The impact of the slope angle is considered by assigning to each slope angle a spacing factor starting from 5 MW per km² for slopes between 0 and 3°, 3 MW per km² for slope angles between 3 and 5° and 2.5 MW per km² for slope angles between 5 and 10° (K. David et al, 2008).

4.7 GIS analysis

After analysing the wind speed data and obtaining energy yields at each given coordinate based on the previously stated parameters, a spatial interpolation technique is used to predict the wind speed in locations where data are not available. A variety of deterministic and geo-statistical methods are available to interpolate the values of meteorological phenomena.

In this analysis, the method used to convert the point data into raster format is the Inverse distance weighted (IDW). IDW interpolates a raster surface from points using an inverse distance weighted technique (ESRI, 2011). It determines cell values using a linearly weighted combination of a set of sample points. The weight is a function of inverse distance. The surface being interpolated should be that of a locationally dependent variable. IDW is a technique used to estimate variables, such as wind speed in some points of a spatial field starting from known values sampled in other points (M.Cellura et al, 2007). In this analysis the variables that depend on the location are the yearly energy yield, the mean wind speed and the capacity factor. This method assumes that the variable being mapped decreases in influence with distance from its sampled location. The characteristics of the interpolated surface are controlled by limiting the input points used in the calculation of each output cell value (4points in this analysis). This is also done because input points far away from the cell location where the prediction is being made may have poor or no spatial correlation, so there may be reason to eliminate them from the calculation.

At this point it is essential to illustrate how IDW can be expressed mathematically (G.Xyids et al, 2009). Let us consider n sites in which the variables Z_i are measured. The value Z_p in the station P (called pivot station) can be estimated by using a weighted mean of the available measurements as the formula below shows:

$$Z_P = \frac{\sum_{i=1}^n W(r_{i,P}) Z_i}{\sum_{i=1}^n W(r_{i,P})}$$

where $r_{i,P}$ is the distance between the i -th measurement station and the pivot point P and $W(r_{i,P})$ the weighting factor, which is a decreasing function of the distance. Its value decreases with the distance following a quadratic or an exponential law.

A geo-statistical method is sufficient if data exhibit a Gaussian distribution (B. Sliz-Szkliniarz et al, 2010). To verify the interpolation accuracy and validity of the model, a cross-validation tool provided through ArcGIS Geostatistical Analyst is applied. The root mean square (RMS=0.069) and mean error are relatively low, hence the IDW is chosen to proceed. The Kriging method, yet another interpolation method is suggested as future work in order to compare its results with the ones obtained from IDW.

5 Results

In this section, several types of results are presented; technical wind power potential wind speed and capacity factor maps as well as a consistent table. Madagascar and Somalia are excluded from this part of the analysis due to lack of high resolution data and absence of suitable areas. A first insight about the wind power potential in these countries can be provided through an overlay of the wind power density map and the ultimate suitability map. Despite the fact that in Somalia it is experienced high wind power potential the restriction parameters do not allow yet the implementation of large scale wind power installations. In Madagascar there are suitable areas with sufficient wind power potential in the southern part and the northern part and also in some areas along the central Madagascar.

The mean wind speed, the yearly wind energy yield produced at each grid cell (grid size 1km×1km) that can be technically exploited and the corresponding capacity factor are illustrated in the following maps. In the Appendix B there are tables with detailed information about the mapped values and how they are being shaped for the various countries. For instance, there are tables showing the area percentage at each country that experiences a certain capacity factor and a certain annual energy yield. These tables can be valuable since they provide with crucial information regarding the wind power potential strength.

At first a map of wind speeds at 80m hub height in Africa is presented. The several wind speeds are classified in wind speed classes as suggested in previous chapter using a Spatial Analyst tool (*Reclassify*) in ArcGIS. Thereafter, there are two maps demonstrating the technical wind power potential map in Africa one including the distance to grid restriction criterion and another excluding it. Likewise two maps represent the capacity factors along the geographically suitable areas for wind farm installations in Africa.

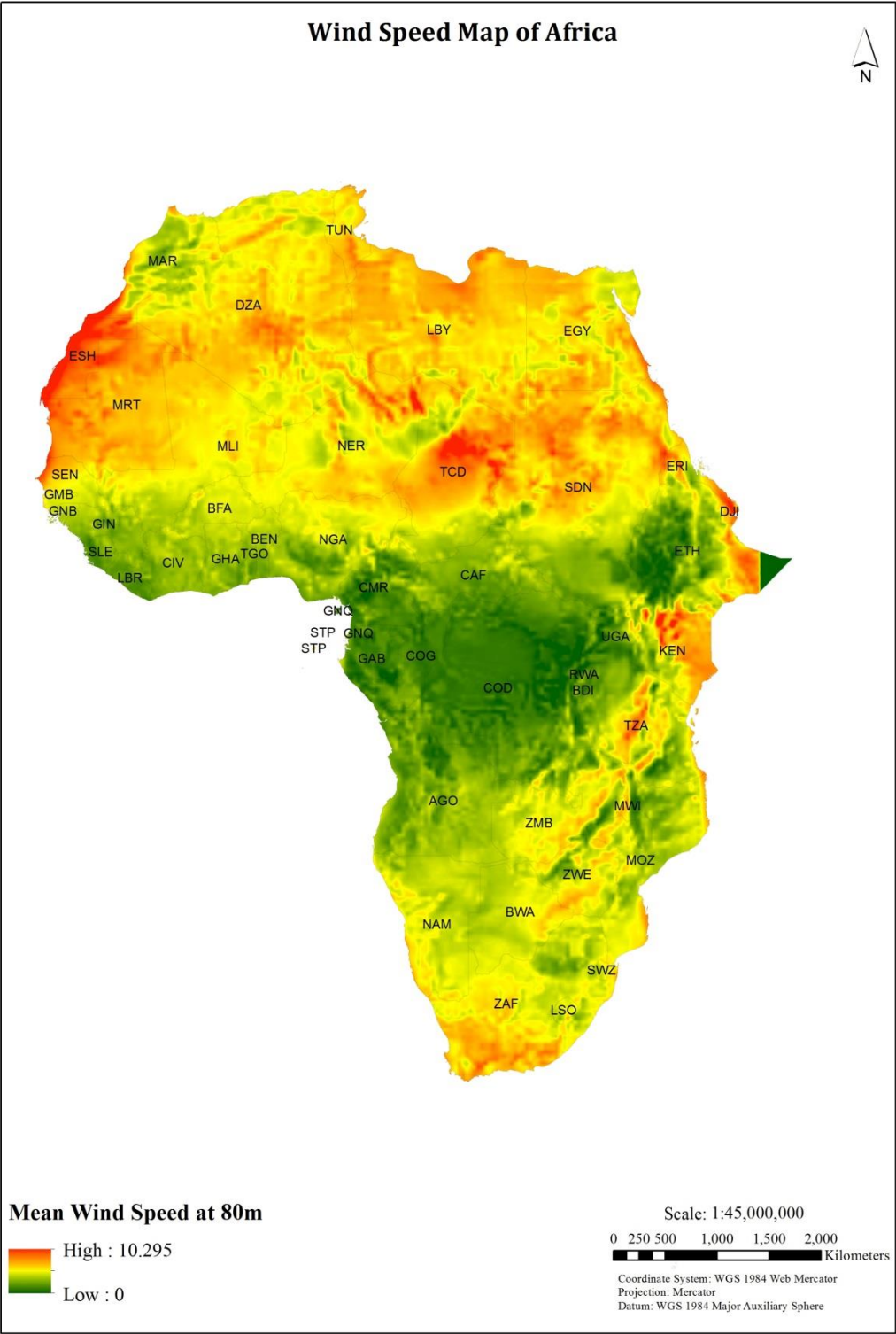


Figure 5.1: Map of wind speeds at 80m hub height in Africa

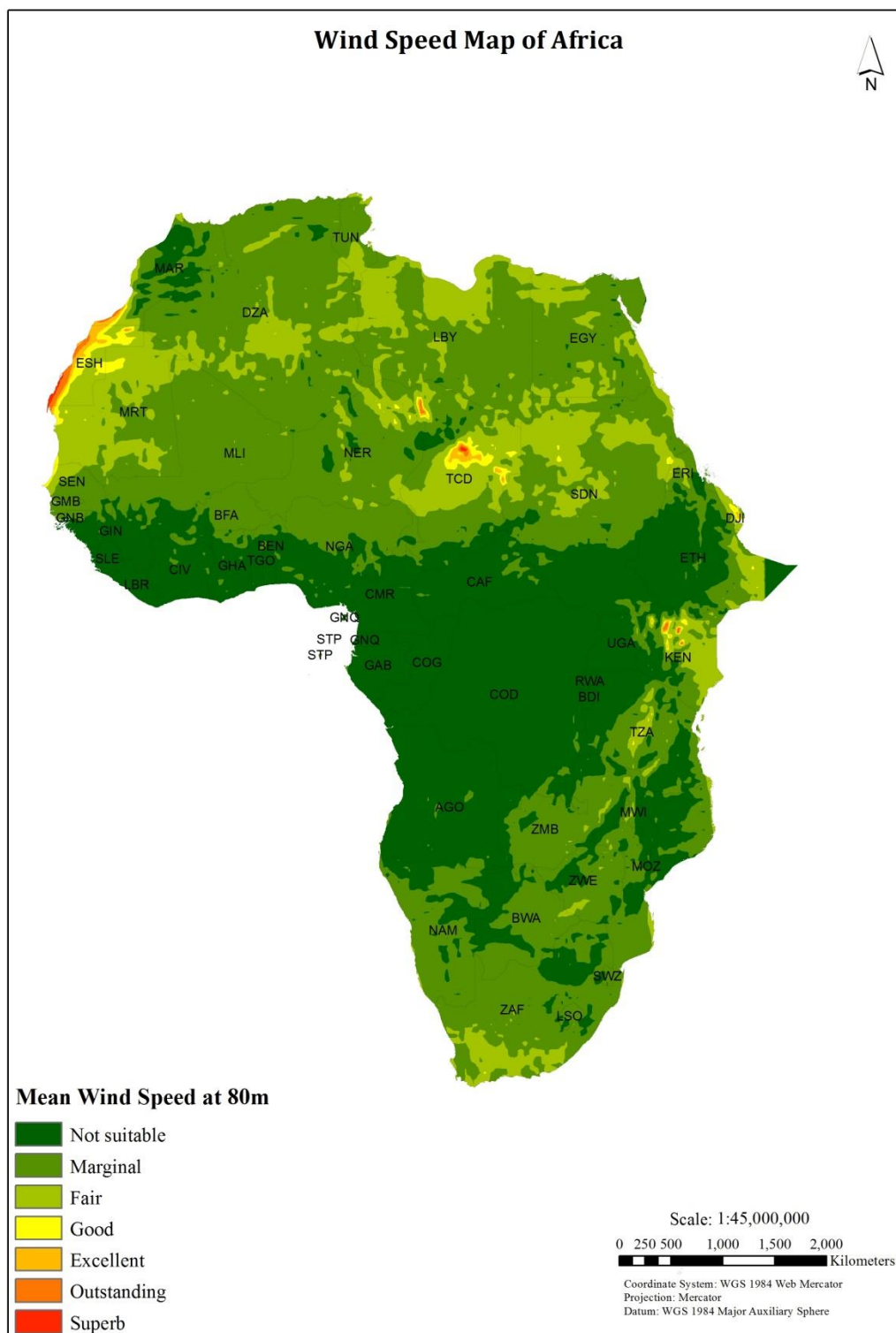


Figure 5.2: Map of wind speeds at 80m hub height in Africa-(classified)

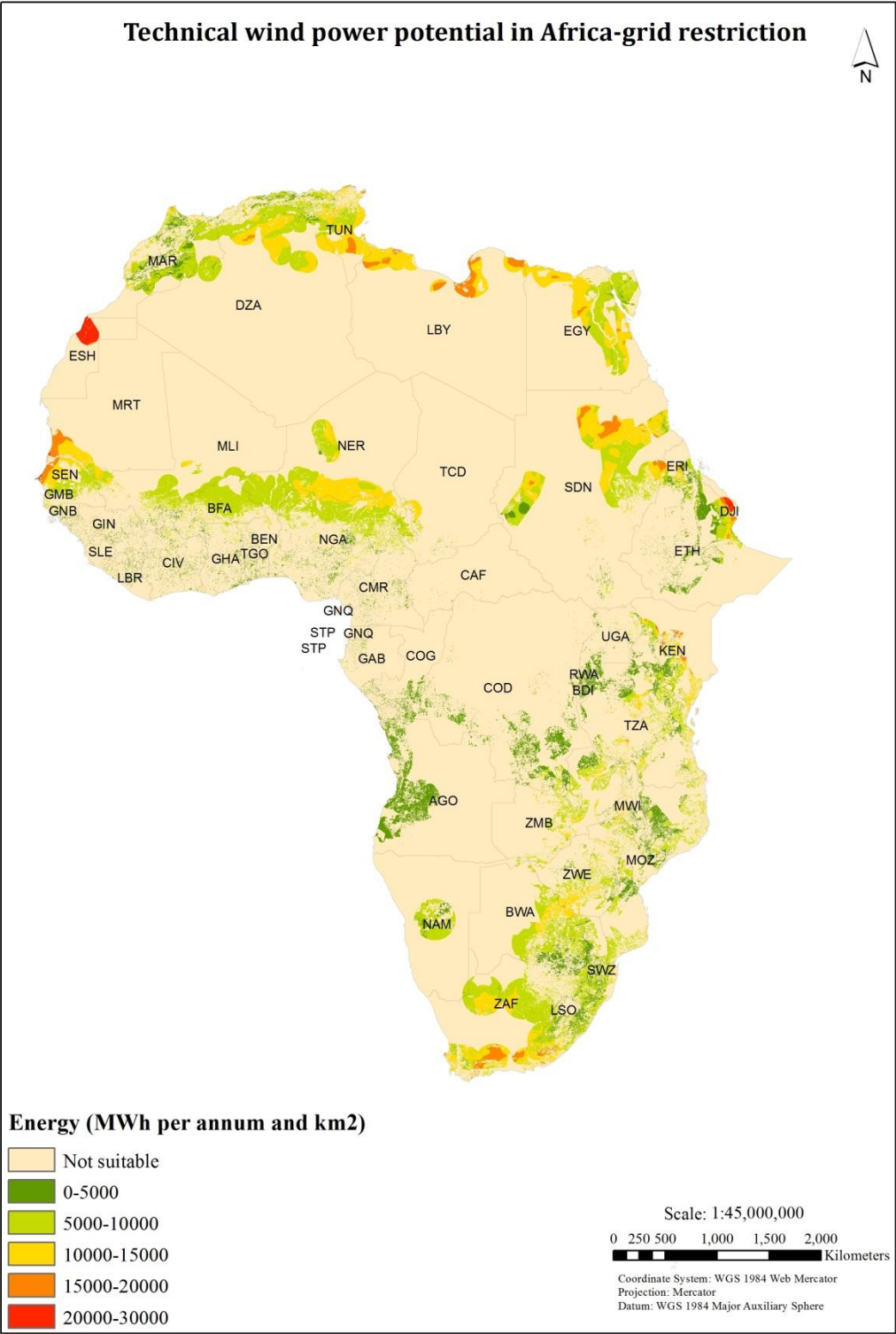


Figure 5.3: Technical wind power potential map in Africa-grid restriction

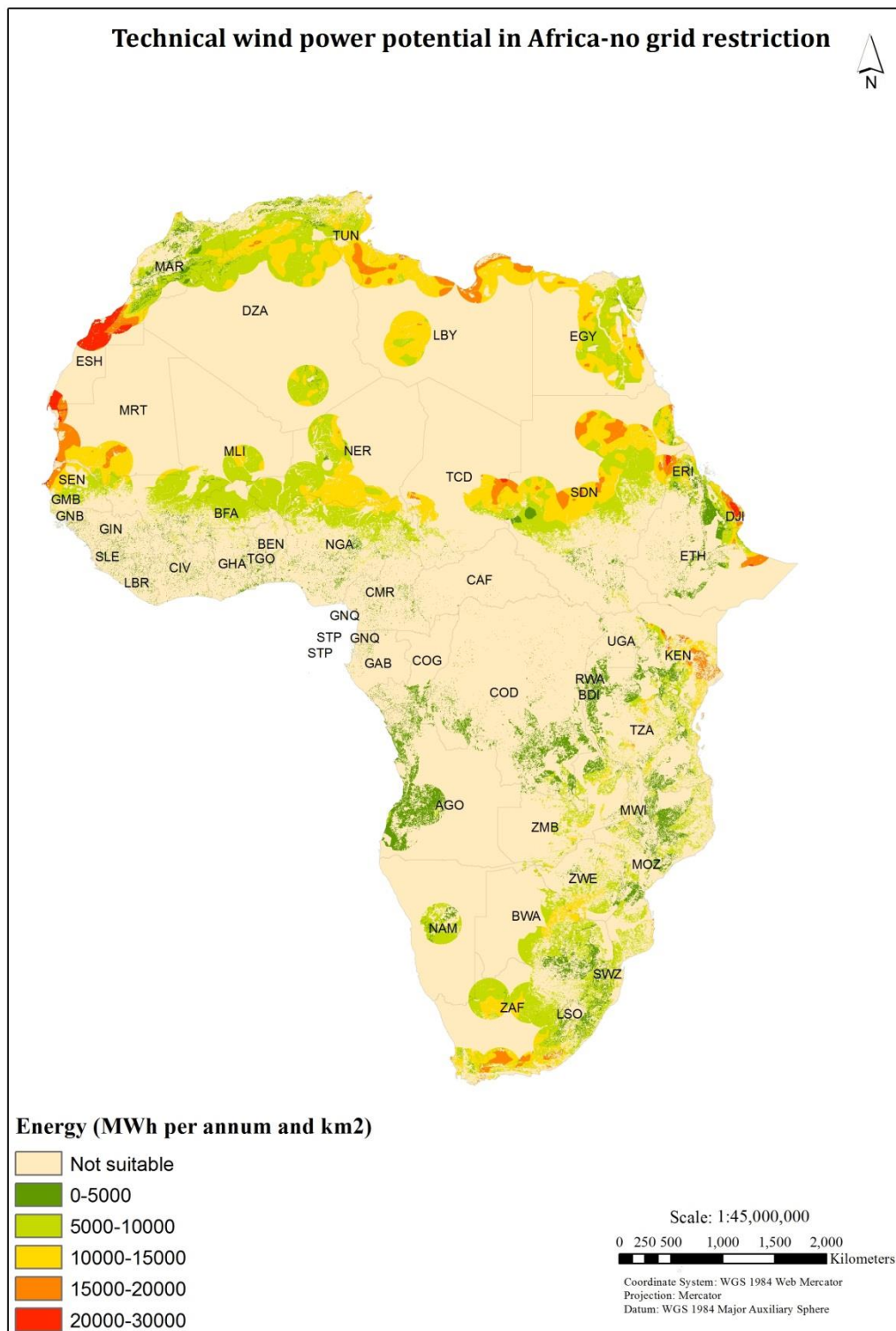


Figure 5.4: Technical wind power potential map in Africa-no grid restriction

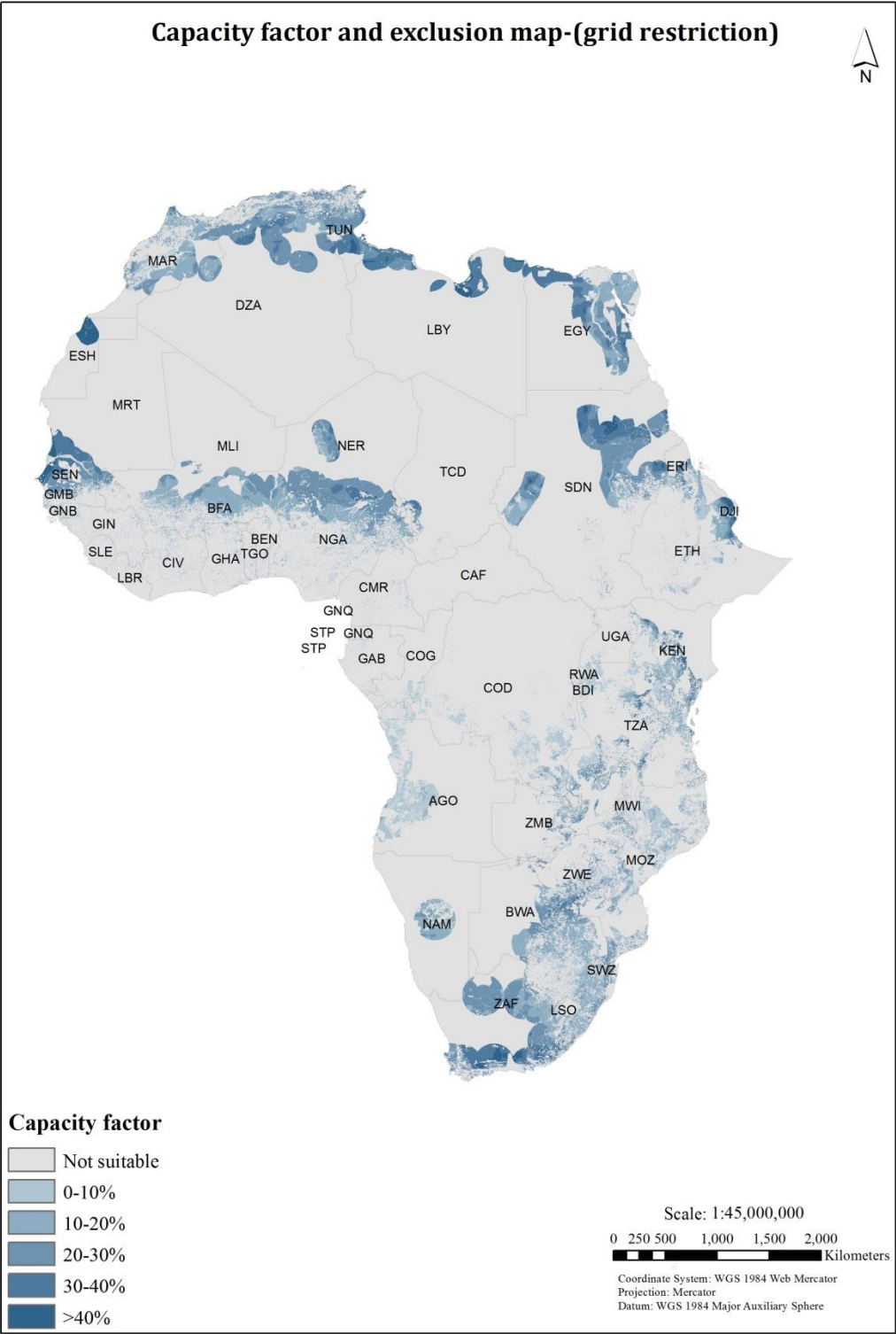


Figure 5.5: Capacity factor map-grid restriction

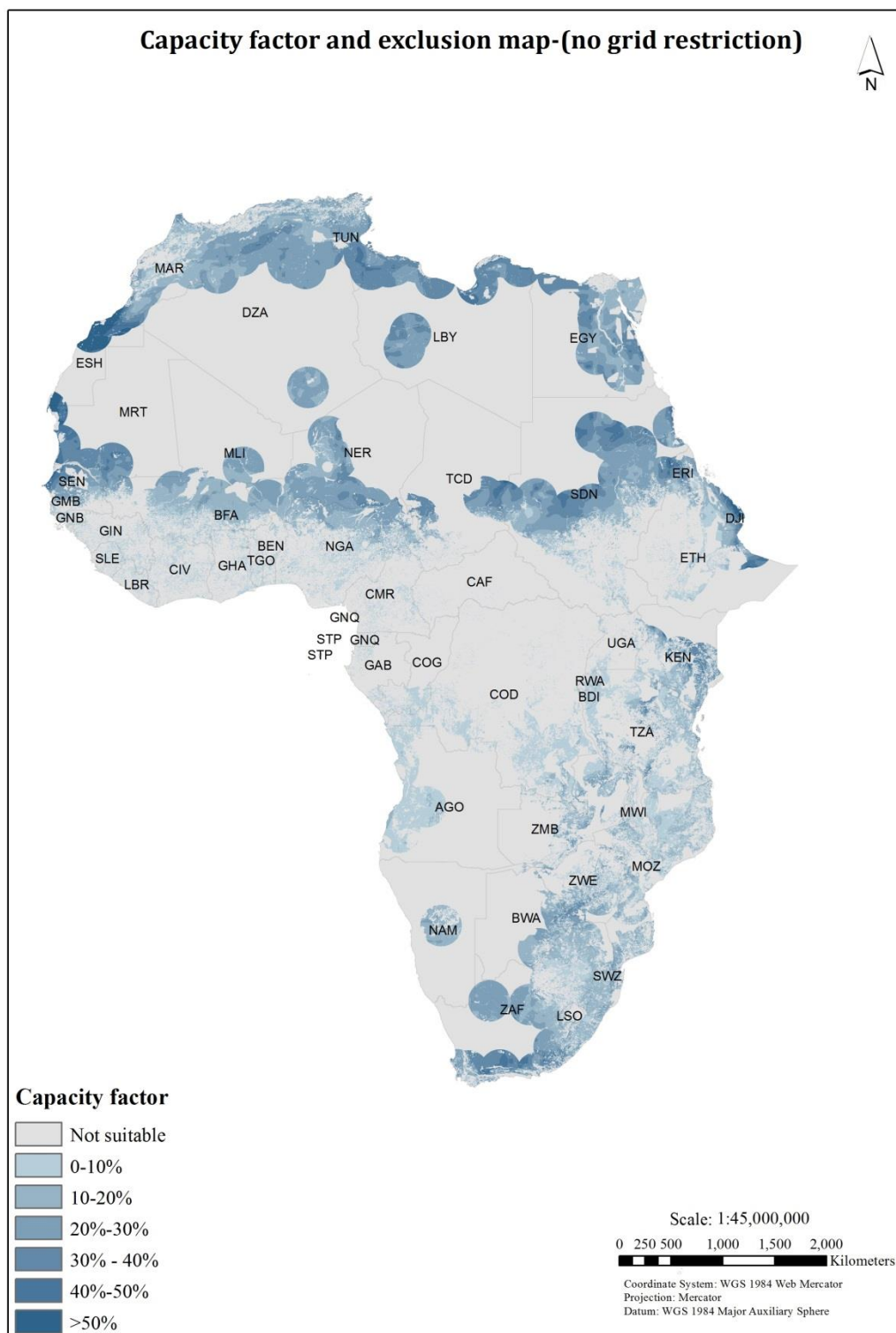


Figure 5.6: Capacity factor map-no grid restriction

The existing and under construction wind farms that are listed in the 2nd chapter are mapped in order to validate the energy assessment carried out in this study. It is apparent that the wind farms that are being installed or are under development stage are located in areas where the technical wind power potential is satisfactory. From the following map, it is also noticeable that there is a tremendous amount of available areas where wind farms could be installed aiming to increase the wind power penetration on the continent. For instance, in Tunisia there is high wind energy yield. However, only a small area has been utilized for wind power installations.

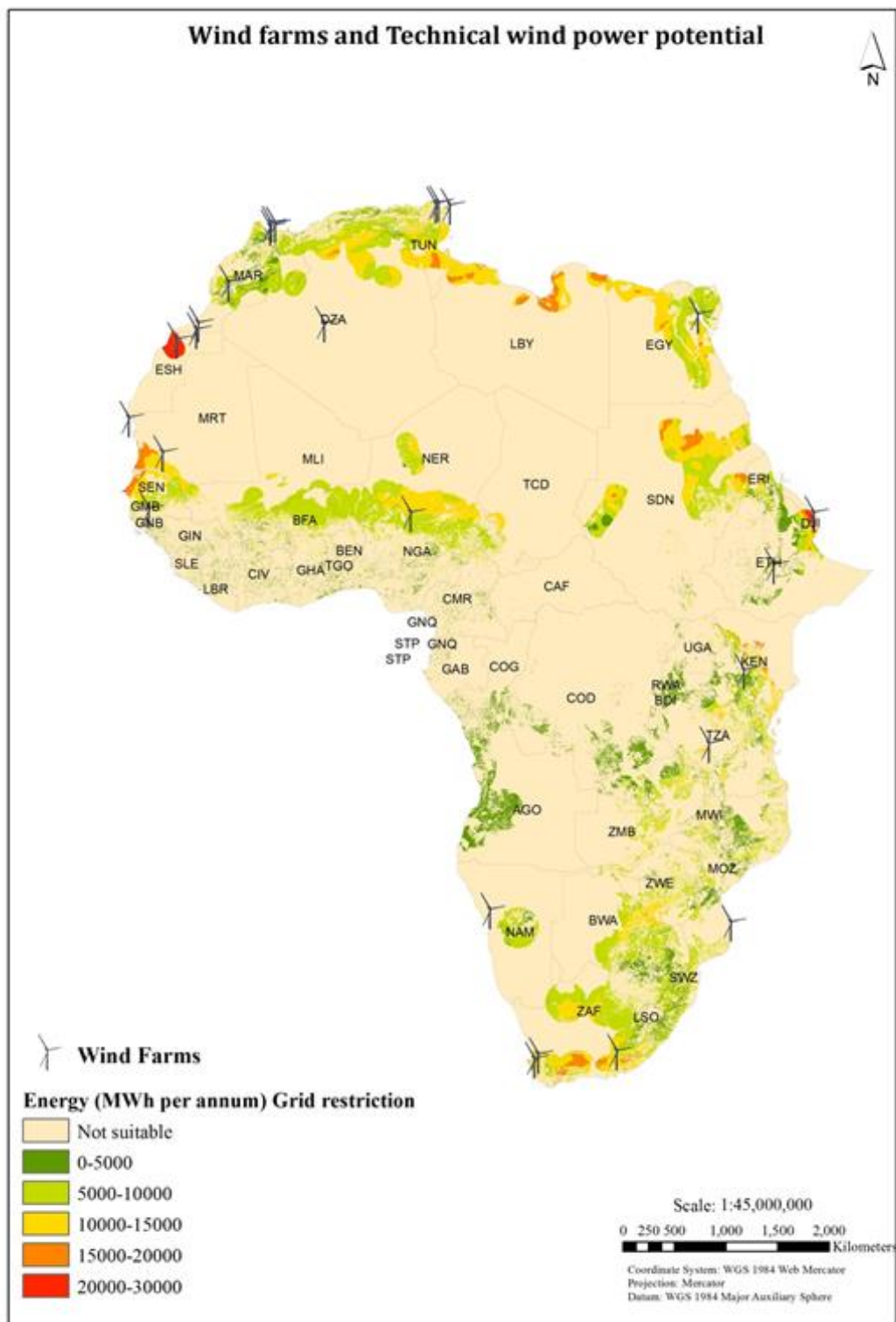


Figure 5.7: Wind farms and wind power potential in Africa

It is essential to illustrate the outcome of this analysis in a comprehensive way. The maps give a decent insight about the technical wind power potential. However, exact data cannot be obtained by observing the map. Thus a spatial analyst tool (*Zonal statistics as a table*) is applied in the technical wind power maps in order to summarize the values these raster have within the zones of Africa's map and report them to a table, so the technical wind power potential can be expressed in TWh/annum for each country. It is apparent that the distance to the grid restriction might have a significant impact on the yearly energy yield. South Africa, Sudan, Algeria, Egypt, Libya and Nigeria signify high yearly wind energy yield.

Areas with capacity factors higher than 20% are selected as potentially suitable for wind power installations that could be economically viable (Bergeles, 2005). The estimation of the economic wind power potential in Africa is not in the scope of this thesis, but it is suggested for future work. Transmission policies and costs, wind farms investment and O&M costs as well as renewables support schemes should be considered. In the following table there are also presented recent data about the total final electricity consumption in African countries so as to perceive figuratively the amount of electricity consumed at the moment and the amount of electricity that could be potentially generated by wind turbines (IEA, 2013; CIA, 2013). It is obvious that the technical wind power potential on the continent exceeds expressively the current electrical power consumption.

Table 5.1: Technical Wind power potential

COUNTRY	Energy (TWh/year) –no grid restriction	Energy (TWh/year) –grid restriction	Energy (TWh/year) CF>20%	Electricity TFC (TWh)
Algeria	7568.9	4021.3	3276.8	33.5
Angola	493.7	421.5	0.0	4.4
Benin	36.7	36.7	0.0	0.8
Botswana	1100.1	1094.7	685.5	3.1
Burkina Faso	938.9	938.9	47.6	0.7
Burundi	15.2	12.1	0.0	0.3
Cameroon	171.0	168.5	30.2	4.9
Central African Republic	7.7	3.8	0.0	0.1
Chad	1977.9	219.5	188.7	0.1
Republic of Congo	41.2	35.9	0.0	0.5
Democratic Republic of the Congo	450.1	378.9	20.4	6.3
Côte d'Ivoire	97.5	97.0	0.0	4.3
Djibouti	260.7	209.9	178.4	0.3
Egypt	5155.9	3560.8	2724.3	125.2
Equatorial Guinea	1.4	1.3	0.0	0.1
Eritrea	992.7	439.4	329.7	0.3
Ethiopia	1159.7	703.5	238.9	3.8
Gabon	6.1	5.8	0.0	1.4
Gambia	43.3	43.3	16.7	0.2
Ghana	82.8	82.8	0.0	6.9
Guinea	60.0	57.0	0.0	0.9

Guinea-Bissau	30.9	30.9	0.1	6.3
Kenya	1582.0	915.8	647.4	5.7
Lesotho	47.6	47.6	5.2	0.2
Liberia	15.4	11.6	0.0	0.3
Libya	6877.4	2575.6	2575.6	23.4
Malawi	89.8	89.8	42.7	1.6
Mali	2479.8	985.5	324.8	0.5
Mauritania	3301.3	1037.3	1033.3	0.5
Morocco	2721.7	1323.7	493.1	23.7
Mozambique	1554.9	1216.0	369.8	10.2
Namibia	580.4	577.0	101.1	3.4
Niger	4581.4	3177.1	2614.2	0.6
Nigeria	2225.1	2219.4	1251.0	20.9
Rwanda	13.9	13.9	0.0	0.2
Sao Tome and Principe	0.6	0.0	0.0	0.0
Senegal	1095.4	1095.4	945.7	2.4
Sierra Leone	18.3	7.9	0.0	0.1
South Africa	6306.7	6040.7	4223.3	213.9
Sudan	11039.2	5432.8	4773.3	6.1
Swaziland	70.7	70.7	31.6	1.2
Tanzania	1261.0	1088.2	513.3	3.7
Togo	22.7	22.7	0.0	0.7
Tunisia	1971.3	1525.0	1476.6	13.8
Uganda	51.9	51.9	12.5	169.9
Western Sahara	2060.3	876.3	876.3	0.1
Zambia	909.5	901.8	475.6	7.8
Zimbabwe	867.6	790.4	542.9	12.5

The above table can be also illustrated in a related graph where the wind energy yield is presented for each country. Equatorial Guinea, Gabon, Central African Republic, Burundi, Liberia, Benin and Togo are countries with lower technical wind power potential that could be technically exploited as shown in the next figure while Sudan, South Africa and Algeria are the leading countries in terms of technical wind power potential.

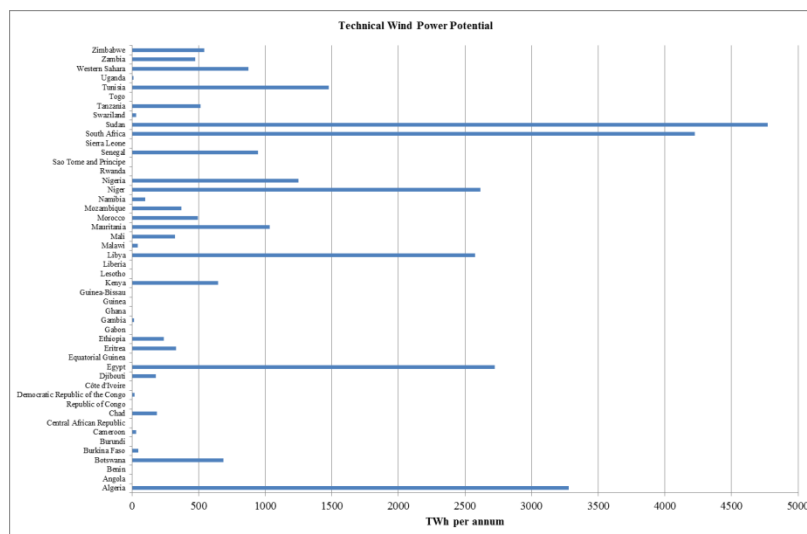


Figure 5.8: Technical wind power potential-African countries

As previously stated there are some countries that indicate high wind energy yield. In the following graph it is depicted how much of the countries areas are available for certain levels of annual wind energy yield (MWh/annum and grid cell 1km²). The countries shown below are Algeria, South Africa, Morocco, Sudan, Egypt, Libya and Nigeria i.e. the countries that indicate particularly great wind power potential. To illustrate, in South Africa about 25% of the country's area is suitable for generating between 5000 and 10000 MWh annually per 1km². Detailed results for all the countries are provided in Appendix Table B.2.

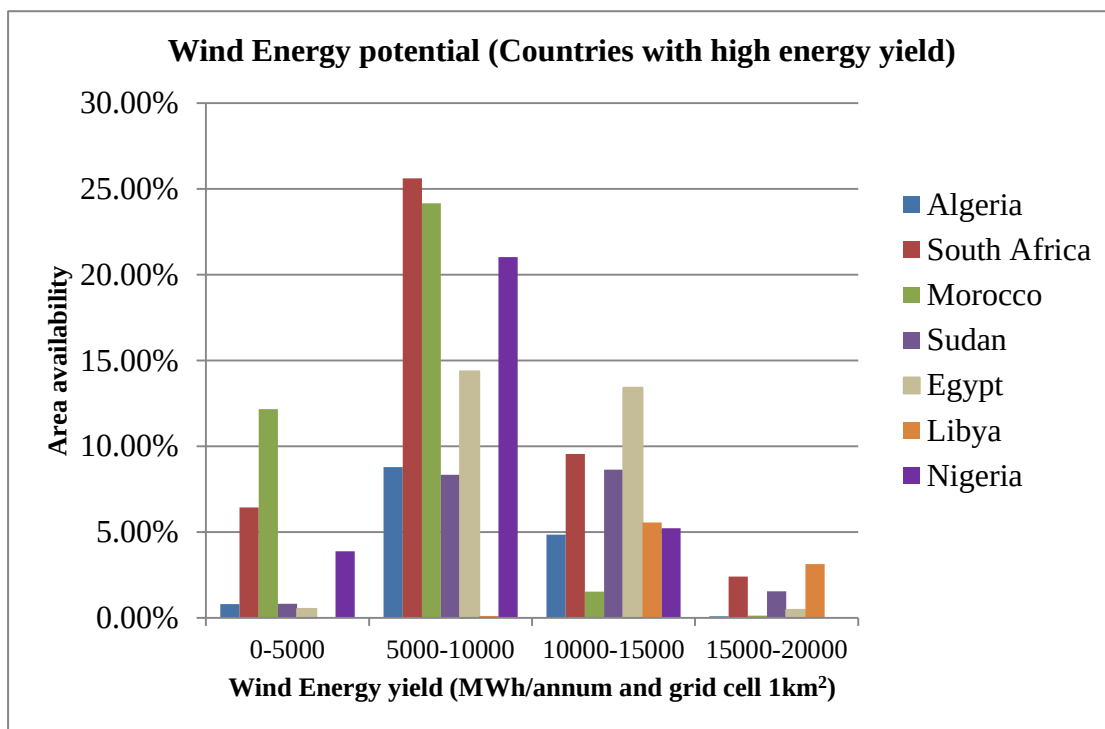


Figure 5.9: Wind Energy yield-Area availability

6 Conclusions and discussion

The theoretical, geographical and technical wind power potential are estimated in this study using 1 year wind speed data, several screening criteria in order to obtain the suitable areas for wind farm installations. Wind turbine power curves, up-to-date spacing and availability factors and array losses are considered. A detailed analysis based on GIS is performed and the results are presented in both maps and tables. All the delivered maps have a certain grid size (1km×1km) and the used projection is Web Mercator WGS1984-Projected Coordinate System. IDW is applied to estimate variables, such as wind speed in some points of a spatial field starting from known values sampled in other points. NASA and Vortex wind speed data are used through this assessment. The height at which the wind speeds are extrapolated is 80m; i.e a modern wind turbine's hub height.

From the analysis, it should be concluded that the wind energy potential on the continent is significant. In most regions, the technical potential exceeds the electricity consumption. The countries with the least wind power potential are Equatorial Guinea, Gabon, Central African Republic, Burundi, Liberia, Benin and Togo while Sudan, South Africa and Algeria are the foremost countries in terms of technical wind power potential.

There are several parameters taken into account that have not been considered in previous studies for the African continent; such are the variation of air density, the relatively small grid size (when referring to a whole continent) for the wind energy assessment, the slope impact and numerous siting criteria applied throughout the whole continent. The default turbine has a rated power $P=2MW$ and the hub height is 80m, therefore the resulted technical wind power potential is higher than the one obtained in (M.Hoogwijk et al, 2004), where the default turbine has half rated power and 69m hub height (just 14% lower than 80m). However, all the figures provided in this study are indicative, since only 1 year's data are available and the data resolution is not as fine for assessing wind power on a fully detailed basis as it would be done in a pre-feasibility study and a specific site assessment.

Although this study follows a qualitative approach, there are some crucial parameters not yet fully quantified, as for instance the impact that the transmission costs would have on the technical potential and the economic viability of wind generated electricity. Wind power markets are quite dependent on the availability of high voltage transmission lines, without which they are not be able to transport large amounts of electricity. This dependence on transmission lines infers that wind energy is particularly sensitive to the cost of transmission. An initial approach on that issue is done by implementing the distance to the grid criterion, but a thorough study should be carried out regarding the optimization of transmission costs and wind power implementation.

To conclude with, the lacking access to electricity is a fact on the continent and Africa should deal with the challenge of generating more power to meet existing and future demand. As demonstrated in this analysis, the continent is endowed with high wind power potential which could penetrate significantly to the continent's energy systems. The findings of this study should be perceived as an insight and a first step to the development of wind energy field in Africa. All the entities involved with energy

systems on the continent, policy makers, turbine manufacturers and wind farm operating and management companies are suggested to use the resulted maps and tables in order to obtain a holistic view of the wind power in Africa and select specific areas for further evaluation. All the data, that are used and analysed in this study, are formed in a compatible to GIS format and stored in a complete database.

7 Future work

Paying respect to the Academia and promoting continuous research activity on the renewable energy assessment field, the following future work is suggested. The economic wind power potential should be assessed based on optimization of transmission costs, investment and O&M costs as well as on renewable energy support schemes. It is highly suggested that more than one year of wind speed measurements should be taken into account and apply the same methodology in order to obtain a more accurate wind speed distribution. Furthermore, the spacing can be more specific by considering the wind speed direction across the continent. By doing so, a model for the whole continent could include not only estimates on wind power potential, but also levelized cost of wind generated electricity, investment indicators (such as Internal Rate of Return, Net Present Value and Pay-Back Period) and indicative wind turbine siting at each grid cell. Additionally, the offshore wind power potential should be estimated in order to obtain a complete overview regarding the extent that wind power can penetrate on the continent. Also the Kriging interpolation method should be applied and its results should be compared with IDW in order to choose the most accurate interpolation method.

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Appendix A

In this Appendix, there are listed technical information and characteristics about the wind turbines.

Table A-1: ENERCON E82-Technical data (Enercon, 2013)

Rated power	2,000 kW
Rotor diameter	82 m
Hub height	80 m
Wind class (IEC)	IECIII A
Turbine concept	Gearbox, variable speed, variable pitch control
Rotor	
Type	Upwind rotor with active pitch control
Direction of rotation	Clockwise
Number of blades	3
Swept area	5,281 m ²
Blade material	Fibreglass (epoxy resin); integrated lightning protection
Rotational speed	Variable, 6 –19.5 rpm
Pitch control	ENERCON blade pitch system, one independent pitching system per rotor blade with allocated emergency supply
Drive train with generator	
Hub	Rigid
Main bearings	Dual-row tapered/single-row cylindrical roller bearings
Generator	ENERCON direct-drive synchronous annular generator
Grid feeding	ENERCON converter
Braking systems	3 independent blade pitch systems with emergency supply, rotor brake, rotor lock
Yaw control	Active via adjustment gears, load-dependent damping
Cut-out wind speed	25 m/s
Remote monitoring	ENERCON SCADA

Table A-2: VESTAS V90-Technical data (Vestas, 2005)

Rated power	2,000 kW
Rotor diameter	90 m
Hub height	80 m
Wind class (IEC)	IEC/NVN II
Turbine concept	Gearless, variable speed, variable pitch control

Rotor	
Type	Upwind rotor with active pitch control
Direction of rotation	Clockwise
Number of blades	3
Swept area	6,362 m ²
Blade material	Fibreglass reinforced epoxy and carbon fibres
Rotational speed	Variable, 9 –14.9 rpm
Pitch control	Vestas blade pitch system, one independent pitching system per rotor blade with allocated emergency supply
Drive train with generator	
Hub	Rigid
Main bearings	Spherical roller bearings
Generator	Asynchronous with wound rotor, slip rings and VCS
Grid feeding	Vestas converter
Braking systems	3 independent blade pitch systems with emergency supply, rotor brake, rotor lock
Yaw control	Plain bearing system with built-in friction
Cut-out wind speed	25 m/s
Remote monitoring	Possibility of connection of serial communication

Appendix B

In this Appendix, there are listed tables providing with detailed results regarding the overall analysis. Table B-1 shows how each parameter affect the geographic wind power potential in each country. Table B-2 lists the technical wind power potential including grid restriction (MWh per annum and km²) of each country, while Table B-3 represents the Capacity factors and corresponding country based area availabilities (also including grid restriction).

Table B-1: Suitability parameters (country based)

	Distance to the grid	Extreme rural areas	Agriculture	Water bodies	Protected areas	Forest cover	Slope	Elevation
COUNTRY	Percentage of area availability	Percentage of area availability	Percentage of area availability	Percentage of area availability	Percentage of area availability	Percentage of area availability	Percentage of area availability	Percentage of area availability
Algeria	19.47%	32.44%	96.26%	98.19%	95.90%	99.61%	99.66%	99.52%
Angola	47.29%	32.00%	76.18%	97.07%	87.66%	58.67%	99.52%	99.88%
Benin	99.83%	99.88%	66.84%	96.70%	76.40%	28.15%	99.87%	99.68%
Botswana	81.01%	32.64%	85.00%	93.45%	63.43%	99.69%	98.92%	100.00%
Burkina Faso	100.00%	95.05%	68.50%	98.26%	85.59%	87.22%	99.94%	100.00%
Burundi	88.86%	100.00%	82.42%	90.49%	96.39%	65.15%	91.74%	92.78%
Cameroon	84.38%	94.58%	91.30%	94.66%	88.37%	17.38%	98.49%	99.71%
Cape Verde	96.16%	92.68%	76.64%	100.00%	97.56%	96.77%	84.88%	99.34%
Central African Republic	41.59%	55.14%	96.22%	96.05%	82.47%	3.09%	99.95%	100.00%
Chad	8.86%	27.75%	88.97%	87.35%	90.54%	91.58%	99.43%	99.37%
Comoros	37.19%	0.00%	83.20%	100.00%	99.26%	17.62%	73.96%	98.08%
Côte d'Ivoire	71.64%	50.74%	96.70%	97.82%	90.10%	13.43%	99.75%	100.00%
Democratic Republic of the Congo	43.52%	78.54%	97.75%	91.99%	89.62%	12.68%	98.60%	99.42%
Djibouti	97.59%	98.75%	87.87%	92.94%	78.06%	15.65%	99.84%	99.89%
Egypt	80.45%	99.79%	98.94%	97.39%	100.00%	99.42%	97.65%	98.79%
Equatorial Guinea	49.90%	53.35%	95.57%	98.65%	83.89%	99.34%	99.24%	96.93%
Eritrea	93.57%	99.83%	94.71%	98.68%	76.70%	12.13%	98.03%	99.78%
Ethiopia	49.34%	93.38%	90.87%	97.62%	94.55%	99.65%	94.56%	94.49%
French Southern Territories	78.37%	69.34%	74.74%	100.00%	79.82%	70.05%	95.45%	83.18%
Gabon	0.00%	0.00%	87.10%	94.07%	100.00%	25.81%	91.67%	100.00%
Gambia	70.80%	34.59%	96.93%	77.53%	84.90%	11.83%	99.80%	100.00%
Ghana	99.83%	99.83%	73.22%	93.83%	96.03%	89.01%	99.83%	99.83%
Guinea	99.98%	99.98%	75.72%	97.92%	85.45%	24.62%	98.82%	99.83%
Guinea-	96.41%	99.98%	87.11%	84.94%	95.32%	12.83%	99.31%	99.95%

Bissau								
Kenya	98.66%	99.48%	91.30%	93.78%	73.23%	34.16%	99.86%	99.91%
Lesotho	58.84%	65.82%	74.15%	98.18%	90.31%	84.18%	97.39%	94.23%
Liberia	100.00%	100.00%	85.07%	99.35%	78.93%	99.78%	93.15%	38.13%
Libya	68.90%	94.00%	93.70%	98.35%	86.89%	15.22%	99.87%	99.99%
Madagascar	14.03%	27.51%	99.36%	97.77%	99.96%	99.98%	99.83%	99.34%
Malawi	48.21%	68.94%	55.37%	75.65%	91.30%	71.95%	98.38%	99.73%
Mali	100.00%	100.00%	49.32%	94.94%	84.52%	71.65%	80.23%	98.82%
Mauritania	26.75%	39.23%	92.53%	97.50%	97.36%	93.29%	99.72%	100.00%
Mayotte	7.31%	21.56%	99.98%	100.00%	99.43%	99.98%	99.62%	99.52%
Morocco	48.40%	0.00%	75.00%	98.51%	98.77%	18.04%	99.47%	100.00%
Mozambique	76.08%	94.73%	72.27%	92.09%	98.23%	99.27%	97.06%	95.00%
Namibia	75.18%	93.59%	56.73%	96.91%	82.90%	83.76%	98.62%	99.96%
Niger	86.04%	14.24%	85.43%	95.68%	57.31%	99.91%	99.59%	99.64%
Nigeria	29.85%	42.80%	99.58%	90.73%	83.83%	99.99%	99.78%	100.00%
Republic of Congo	99.36%	100.00%	74.25%	73.43%	86.87%	57.53%	99.29%	99.93%
Rwanda	100.00%	100.00%	88.73%	88.52%	90.04%	55.58%	95.33%	81.66%
Sao Tome and Principe	0.00%	97.93%	92.94%	100.00%	69.99%	27.92%	89.11%	100.00%
Senegal	99.70%	99.99%	87.23%	92.82%	74.68%	81.54%	99.61%	99.59%
Seychelles	0.00%	0.00%	95.24%	100.00%	50.24%	20.63%	81.40%	97.67%
Sierra Leone	45.97%	99.93%	97.18%	89.87%	94.93%	13.52%	99.62%	99.91%
Somalia	1.18%	34.56%	82.01%	96.93%	99.77%	97.78%	99.32%	99.91%
South Africa	93.91%	73.72%	81.41%	98.77%	93.11%	94.18%	98.98%	99.48%
Sudan	39.16%	67.02%	86.52%	96.34%	98.25%	78.47%	99.39%	99.95%
Swaziland	100.00%	100.00%	69.65%	99.82%	97.08%	89.56%	98.93%	100.00%
Tanzania	78.32%	99.31%	54.48%	89.05%	67.79%	83.62%	93.44%	98.54%
Togo	99.99%	99.99%	78.32%	96.29%	89.45%	29.73%	99.70%	99.90%
Tunisia	83.97%	97.42%	84.00%	92.45%	99.71%	99.13%	99.71%	98.61%
Uganda	99.27%	98.37%	94.05%	76.68%	85.12%	14.84%	86.31%	98.33%
Western Sahara	11.67%	29.49%	100.00%	98.42%	94.57%	99.96%	99.90%	99.91%
Zambia	82.82%	62.98%	57.39%	90.37%	64.62%	80.00%	98.57%	99.99%
Zimbabwe	93.79%	95.66%	36.10%	98.95%	72.31%	97.78%	99.23%	99.94%

Table B-2: Wind Energy potential (Technical Potential) including grid restriction (MWh per annum and km²)

COUNTRY	0	0-5000	5000-10000	10000-15000	15000-20000	20000-25000	25000-30000
Algeria	85.45%	0.82%	8.78%	4.85%	0.09%	0.00%	0.00%
Angola	87.46%	12.37%	0.16%	0.00%	0.00%	0.00%	0.00%
Benin	92.99%	3.99%	3.01%	0.00%	0.00%	0.00%	0.00%
Botswana	81.11%	0.05%	13.90%	4.95%	0.00%	0.00%	0.00%
Burkina Faso	53.28%	1.55%	45.17%	0.00%	0.00%	0.00%	0.00%
Burundi	63.09%	36.91%	0.00%	0.00%	0.00%	0.00%	0.00%

Cameroon	92.53%	3.47%	3.74%	0.26%	0.00%	0.00%	0.00%
Central African Republic	99.75%	0.25%	0.00%	0.00%	0.00%	0.00%	0.00%
Chad	98.48%	0.09%	0.54%	0.89%	0.00%	0.00%	0.00%
Republic of Congo	94.74%	5.26%	0.00%	0.00%	0.00%	0.00%	0.00%
Democratic Republic of the Congo	93.65%	5.69%	0.66%	0.00%	0.00%	0.00%	0.00%
Côte d'Ivoire	92.31%	6.82%	0.87%	0.00%	0.00%	0.00%	0.00%
Djibouti	25.38%	2.66%	26.97%	21.01%	14.12%	9.84%	0.01%
Egypt	71.23%	0.52%	14.36%	13.42%	0.46%	0.00%	0.00%
Equatorial Guinea	95.29%	4.70%	0.01%	0.00%	0.00%	0.00%	0.00%
Eritrea	65.68%	7.95%	12.77%	6.96%	3.31%	3.33%	0.00%
Ethiopia	88.16%	7.01%	3.61%	0.97%	0.25%	0.00%	0.00%
Gabon	97.93%	2.05%	0.02%	0.00%	0.00%	0.00%	0.00%
Gambia	53.35%	0.00%	46.65%	0.00%	0.00%	0.00%	0.00%
Ghana	91.55%	6.12%	2.33%	0.00%	0.00%	0.00%	0.00%
Guinea	93.59%	5.88%	0.53%	0.00%	0.00%	0.00%	0.00%
Guinea-Bissau	84.52%	1.57%	13.91%	0.00%	0.00%	0.00%	0.00%
Kenya	81.90%	4.97%	6.54%	5.04%	1.38%	0.17%	0.01%
Lesotho	75.75%	12.50%	11.72%	0.03%	0.00%	0.00%	0.00%
Liberia	94.16%	5.84%	0.00%	0.00%	0.00%	0.00%	0.00%
Libya	91.18%	0.00%	0.12%	5.56%	3.14%	0.00%	0.00%
Malawi	86.01%	8.68%	3.95%	1.35%	0.00%	0.00%	0.00%
Mali	90.21%	0.56%	8.52%	0.71%	0.00%	0.00%	0.00%
Mauritania	93.65%	0.00%	0.60%	3.78%	1.92%	0.05%	0.00%
Morocco	61.95%	12.17%	24.17%	1.53%	0.13%	0.00%	0.05%
Mozambique	76.83%	9.23%	12.94%	0.98%	0.03%	0.00%	0.00%
Namibia	90.85%	1.25%	7.90%	0.00%	0.00%	0.00%	0.00%
Niger	74.67%	0.08%	15.00%	10.24%	0.01%	0.00%	0.00%
Nigeria	69.86%	3.89%	21.02%	5.23%	0.00%	0.00%	0.00%
Rwanda	62.97%	37.03%	0.00%	0.00%	0.00%	0.00%	0.00%
Sao Tome and Principe	100.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Senegal	53.49%	0.00%	22.73%	17.16%	6.03%	0.58%	0.00%
Sierra Leone	96.19%	3.81%	0.00%	0.00%	0.00%	0.00%	0.00%
South Africa	55.99%	6.44%	25.61%	9.56%	2.40%	0.00%	0.00%
Sudan	80.66%	0.82%	8.34%	8.64%	1.55%	0.00%	0.00%
Swaziland	43.84%	20.85%	35.32%	0.00%	0.00%	0.00%	0.00%
Tanzania	81.89%	7.11%	8.26%	2.54%	0.21%	0.00%	0.00%
Togo	88.33%	10.30%	1.37%	0.00%	0.00%	0.00%	0.00%
Tunisia	41.04%	0.51%	16.83%	36.00%	5.62%	0.00%	0.00%
Uganda	93.96%	4.37%	1.51%	0.16%	0.00%	0.00%	0.00%
Western Sahara	88.66%	0.00%	0.00%	0.00%	0.28%	7.58%	3.48%
Zambia	84.56%	2.89%	10.42%	2.14%	0.00%	0.00%	0.00%
Zimbabwe	79.06%	2.34%	12.00%	6.55%	0.05%	0.00%	0.00%

Table B-3: Capacity factors and corresponding area availabilities-including grid restriction

COUNTRY including grid restriction	0.00	0-0.1	0.1-0.2	0.2-0.3	0.3-0.4	0.4-0.5	0.5-1
Algeria	85.43%	0.00%	3.54%	9.53%	1.44%	0.03%	0.00%
Angola	87.44%	11.38%	1.15%	0.00%	0.00%	0.00%	0.00%
Benin	92.86%	2.95%	4.05%	0.00%	0.00%	0.00%	0.00%
Botswana	81.11%	0.00%	8.90%	8.84%	1.15%	0.00%	0.00%
Burkina Faso	53.28%	0.07%	44.57%	2.08%	0.00%	0.00%	0.00%
Burundi	63.09%	36.91%	0.00%	0.00%	0.00%	0.00%	0.00%
Cameroon	92.46%	3.09%	3.75%	0.40%	0.22%	0.00%	0.00%
Central African Republic	99.75%	0.25%	0.00%	0.00%	0.00%	0.00%	0.00%
Chad	98.48%	0.03%	0.36%	0.58%	0.55%	0.00%	0.00%
Republic of Congo	94.72%	5.26%	0.00%	0.00%	0.00%	0.00%	0.00%
Democratic Republic of the Congo	93.65%	5.37%	0.86%	0.12%	0.00%	0.00%	0.00%
Côte d'Ivoire	92.22%	5.10%	2.58%	0.00%	0.00%	0.00%	0.00%
Djibouti	25.23%	0.18%	19.14%	15.46%	24.52%	9.75%	5.13%
Egypt	71.13%	0.00%	9.77%	9.56%	9.30%	0.12%	0.00%
Equatorial Guinea	94.93%	4.51%	0.18%	0.00%	0.00%	0.00%	0.00%
Eritrea	65.25%	3.24%	12.51%	8.09%	5.68%	2.21%	2.38%
Ethiopia	88.15%	4.79%	5.17%	1.30%	0.47%	0.11%	0.00%
Gabon	97.79%	2.02%	0.04%	0.00%	0.00%	0.00%	0.00%
Gambia	52.73%	0.00%	29.23%	16.87%	0.00%	0.00%	0.00%
Ghana	91.35%	4.44%	3.99%	0.00%	0.00%	0.00%	0.00%
Guinea	93.44%	4.95%	1.45%	0.00%	0.00%	0.00%	0.00%
Guinea-Bissau	82.00%	0.00%	14.98%	0.03%	0.00%	0.00%	0.00%
Kenya	81.85%	1.86%	7.20%	3.91%	4.27%	0.76%	0.10%
Lesotho	75.75%	0.00%	22.14%	2.11%	0.00%	0.00%	0.00%
Liberia	94.08%	5.76%	0.08%	0.00%	0.00%	0.00%	0.00%
Libya	91.14%	0.00%	0.00%	1.29%	6.48%	1.05%	0.00%
Malawi	86.01%	5.95%	4.25%	3.30%	0.49%	0.00%	0.00%
Mali	90.21%	0.03%	7.22%	2.51%	0.02%	0.00%	0.00%
Mauritania	93.63%	0.00%	0.04%	1.03%	4.52%	0.67%	0.09%
Morocco	61.83%	1.94%	25.21%	9.45%	1.16%	0.15%	0.05%
Mozambique	76.63%	4.17%	14.24%	4.58%	0.11%	0.00%	0.00%
Namibia	90.81%	0.22%	7.58%	1.34%	0.00%	0.00%	0.00%
Niger	74.67%	0.00%	5.79%	17.49%	2.03%	0.02%	0.00%
Nigeria	69.73%	2.64%	14.01%	12.54%	0.89%	0.00%	0.00%
Rwanda	62.97%	37.03%	0.00%	0.00%	0.00%	0.00%	0.00%
Sao Tome and Principe	92.19%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Senegal	53.35%	0.00%	9.20%	21.27%	10.50%	4.33%	1.09%
Sierra Leone	95.89%	3.61%	0.19%	0.00%	0.00%	0.00%	0.00%
South Africa	55.94%	1.80%	17.13%	16.98%	6.24%	1.81%	0.01%
Sudan	80.65%	0.33%	3.41%	9.88%	4.87%	0.85%	0.00%
Swaziland	43.84%	0.00%	36.87%	19.30%	0.00%	0.00%	0.00%
Tanzania	81.82%	4.59%	8.24%	3.48%	1.45%	0.32%	0.02%
Togo	88.25%	8.87%	2.78%	0.00%	0.00%	0.00%	0.00%

Tunisia	40.86%	0.00%	2.64%	33.01%	19.48%	3.56%	0.00%
Uganda	93.96%	4.02%	1.46%	0.49%	0.07%	0.00%	0.00%
Western Sahara	88.60%	0.00%	0.00%	0.00%	0.00%	0.56%	10.77%
Zambia	84.56%	2.01%	7.02%	5.80%	0.61%	0.00%	0.00%
Zimbabwe	79.06%	1.06%	8.07%	9.03%	2.77%	0.00%	0.00%